

Introduction to RAG: Techniques and Frameworks

12 / 02 / 2025

 Google Developer Groups

By Mohammed Arbi Nsibi



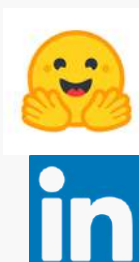
Hello everynryan



MOHAMED ARBI NSIBI

- Final year ICT engineering student@ SUP'COM
- GDG Carthage member
- Mentor of GDGoC SUP'COM & ISAMM
- Former GDSC Lead 23/24

mohammedarbinsibi@gmail.com



<https://huggingface.co/Goodnight7>



<https://www.linkedin.com/in/mohammed-arbi-nsibi-584a43241/>

Content

- Why we need RAG
- RAG components
- Frameworks
- Speaking on Your Behalf : Building a ChatBot
- QUIZ



Why do we need RAG?

By Mohammed Arbi Nsibi

Hallucinations



Is 9677 a prime number?



No, 9677 is not a prime number.
It can be factored into 13 and 745, as $9677 = 13 \times 745$.

} incorrect assertion
} snowballed hallucination

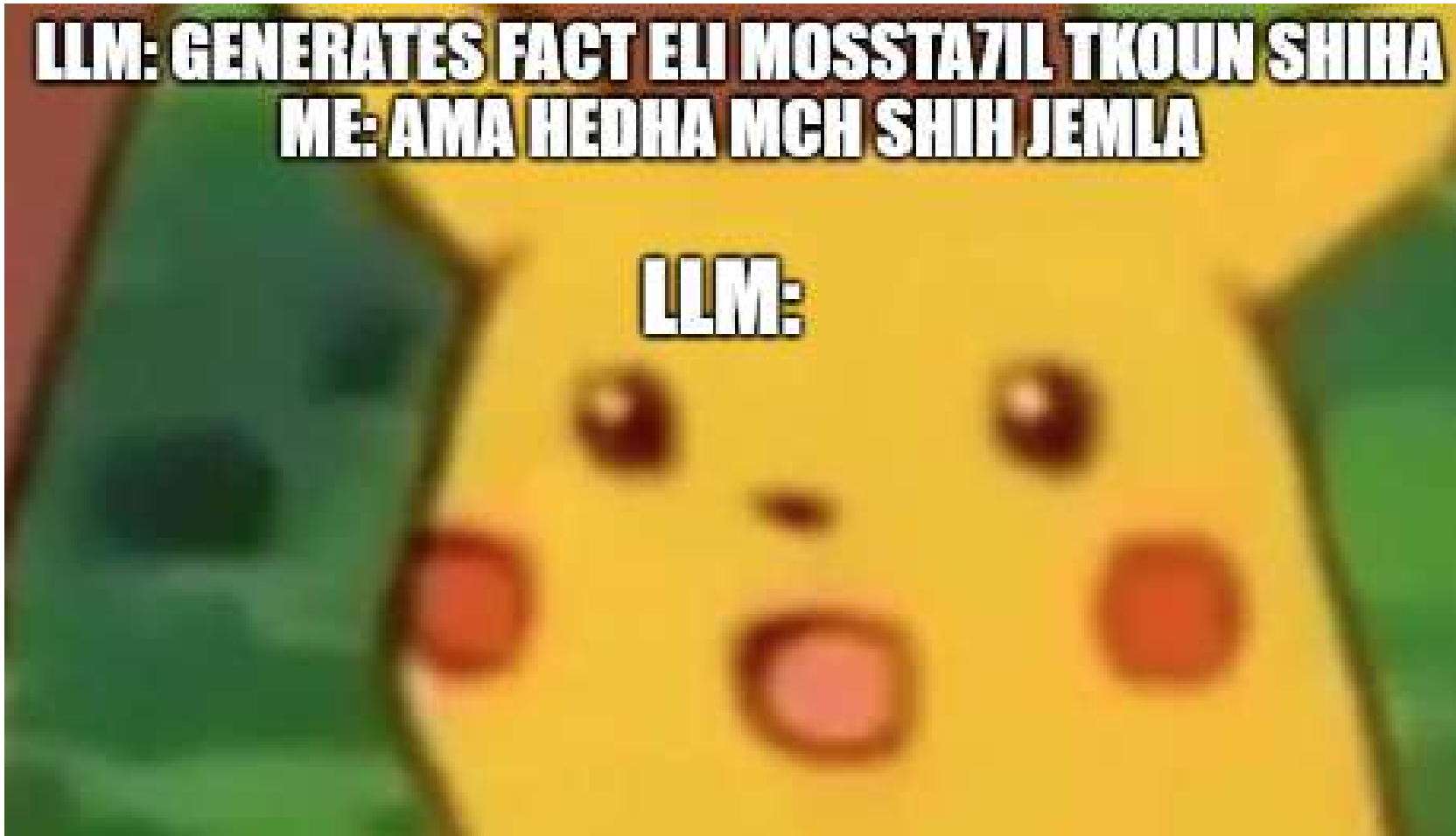
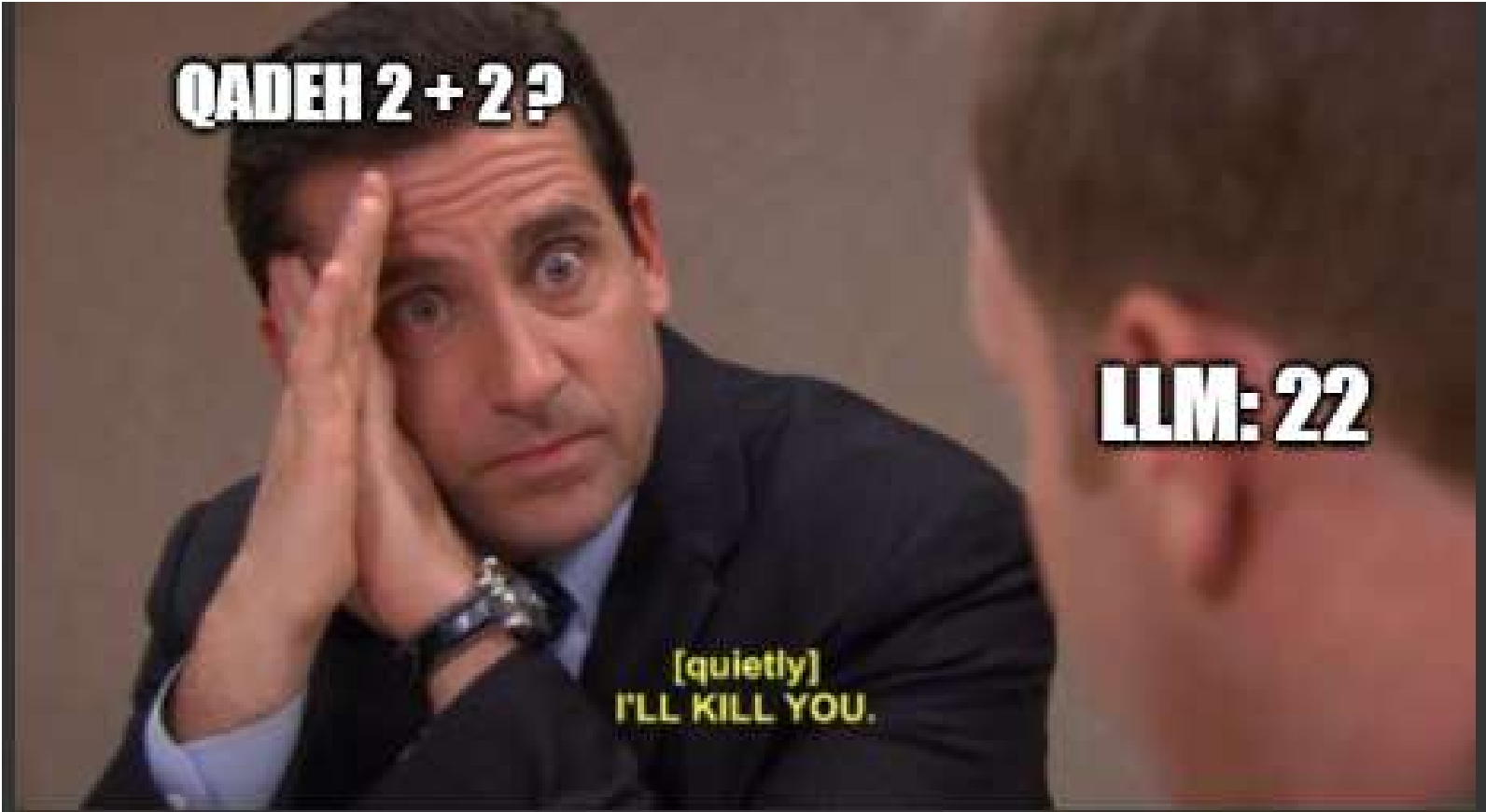


Is 9677 divisible by 13?



No

in a separate session, GPT-4 recognizes its claim as incorrect!



Hallucinations

- The model is not trained on enough data.
- The model is trained on noisy or dirty data.
- The model is not given enough context .
- The model is not given enough constraints (rules, guidelines, or limitations)

LLM AFTER TRAINING
ON 90% OF THE INTERNET...



By Mohammed Arbi Nsibi

Solutions ?

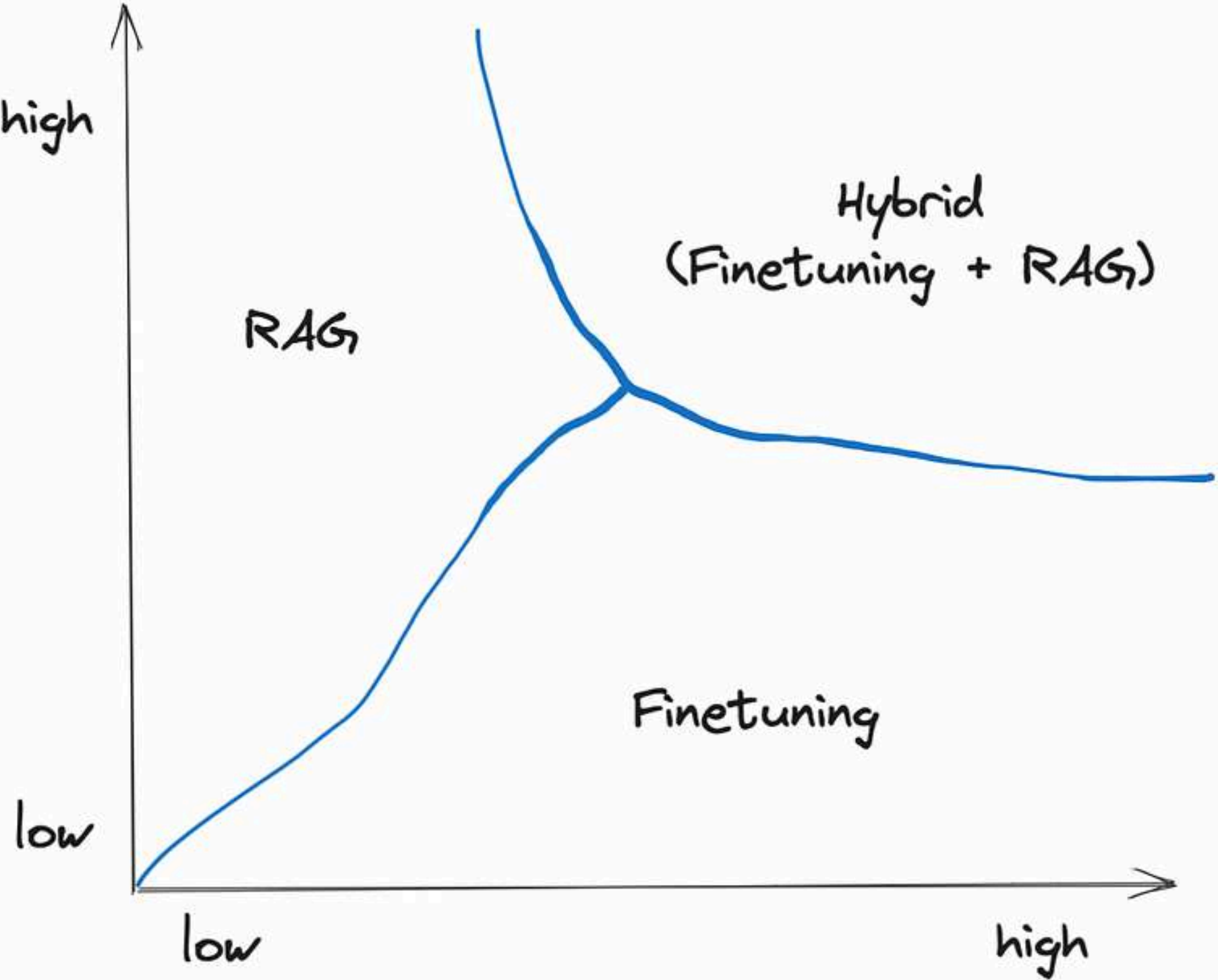


By Mohammed Arbi Nsibi

RAG / Fine-tuning



external knowledge
required



model adaptation required
(e.g. behaviour/
writing style/
vocabulary)

**But wait..
WTF is RAG ?**

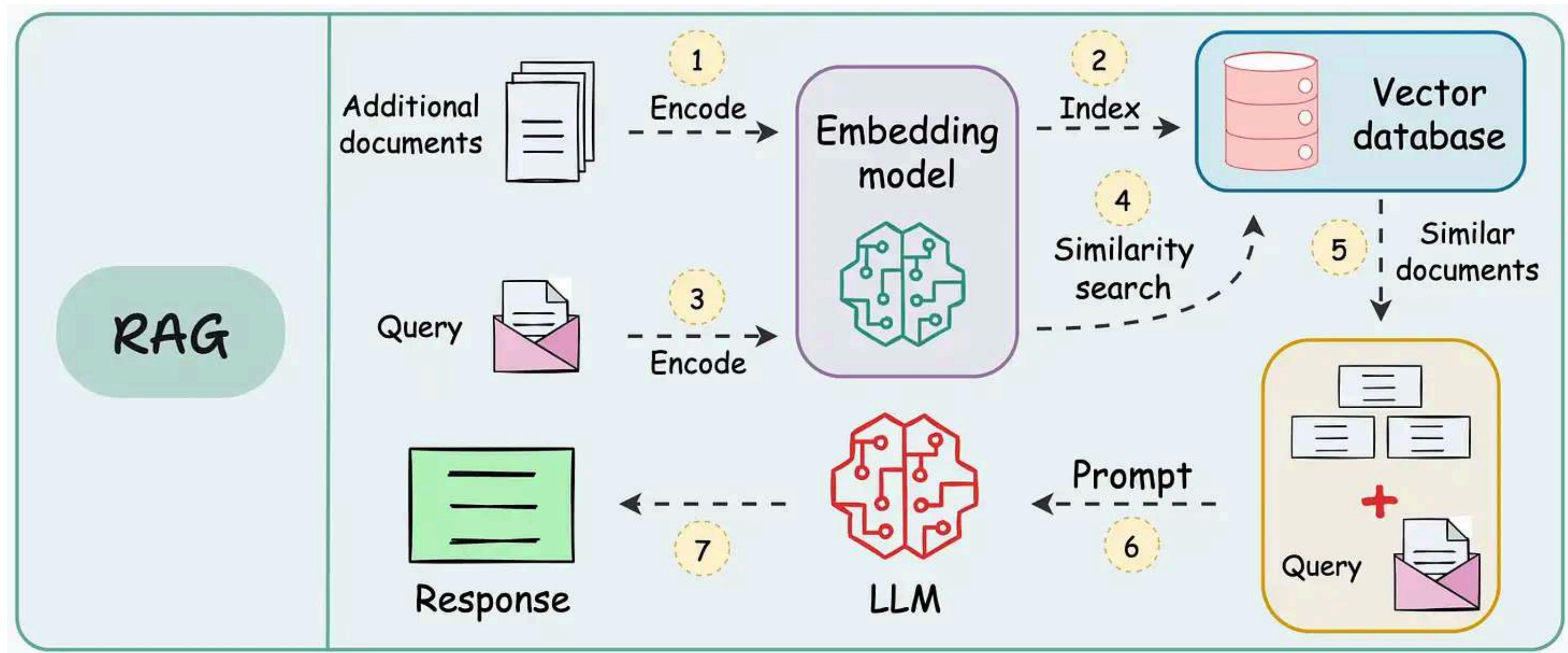


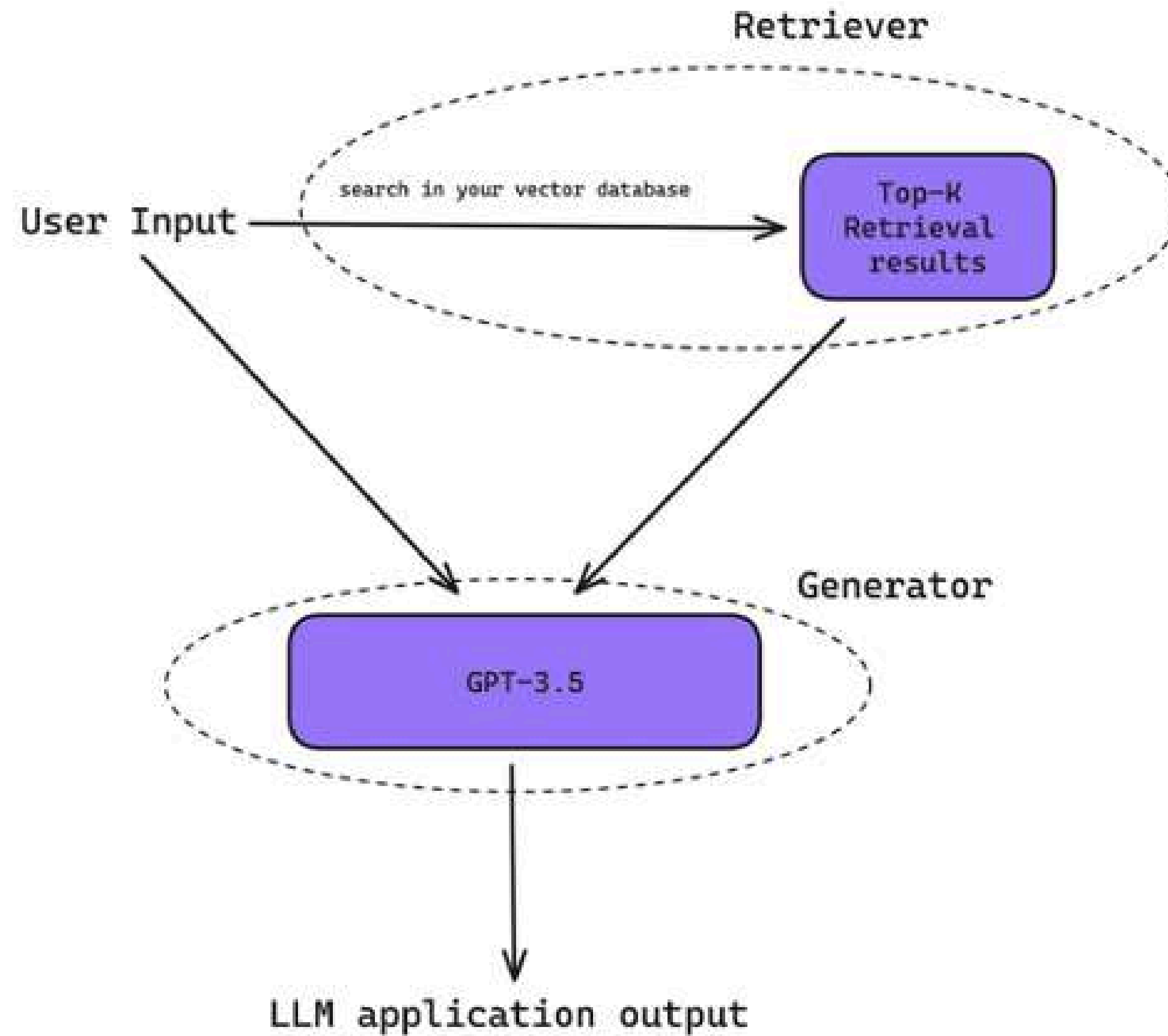
RAG (Retrieval-augmented generation)

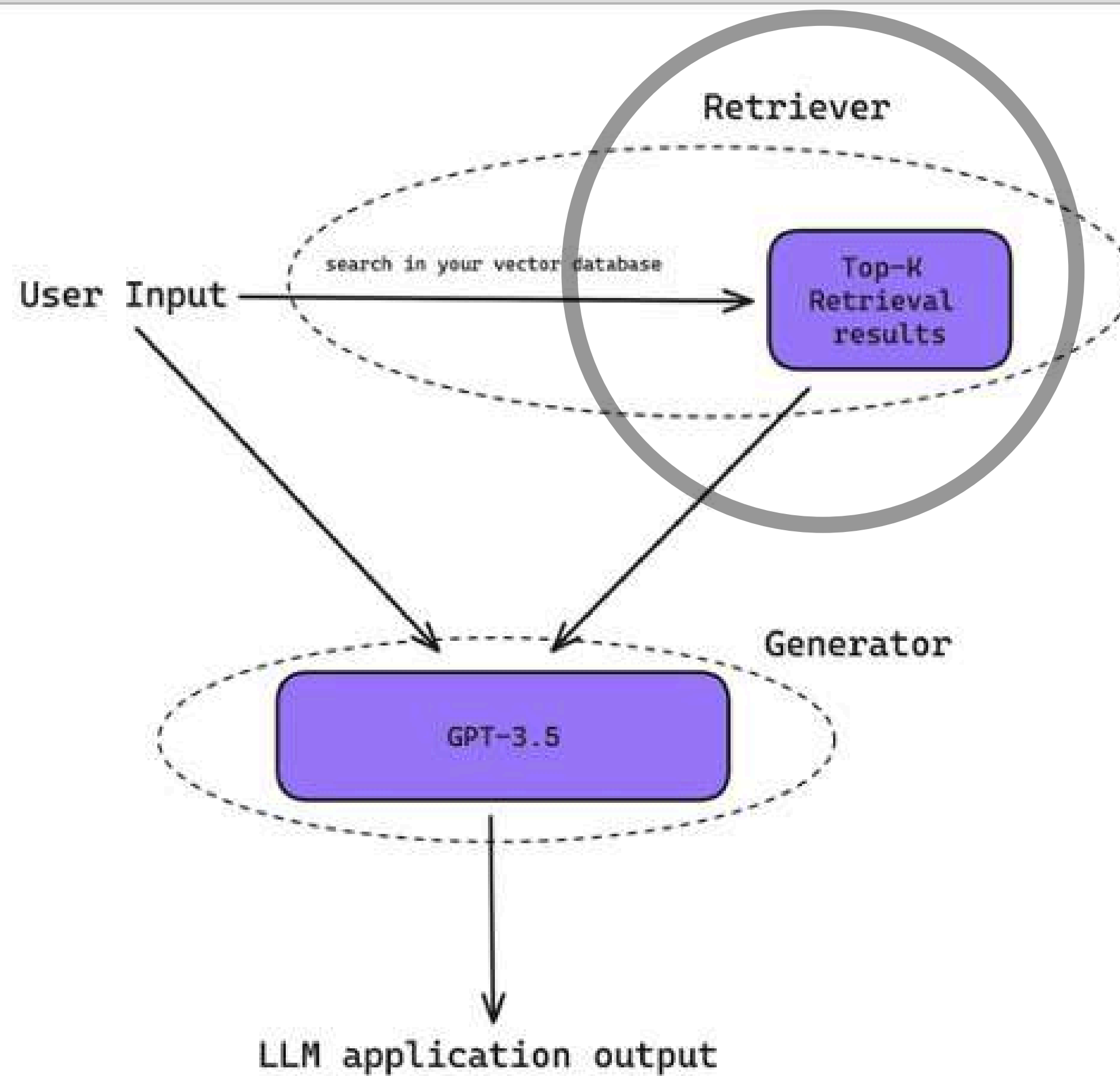


By Mohammed Arbi Nsibi

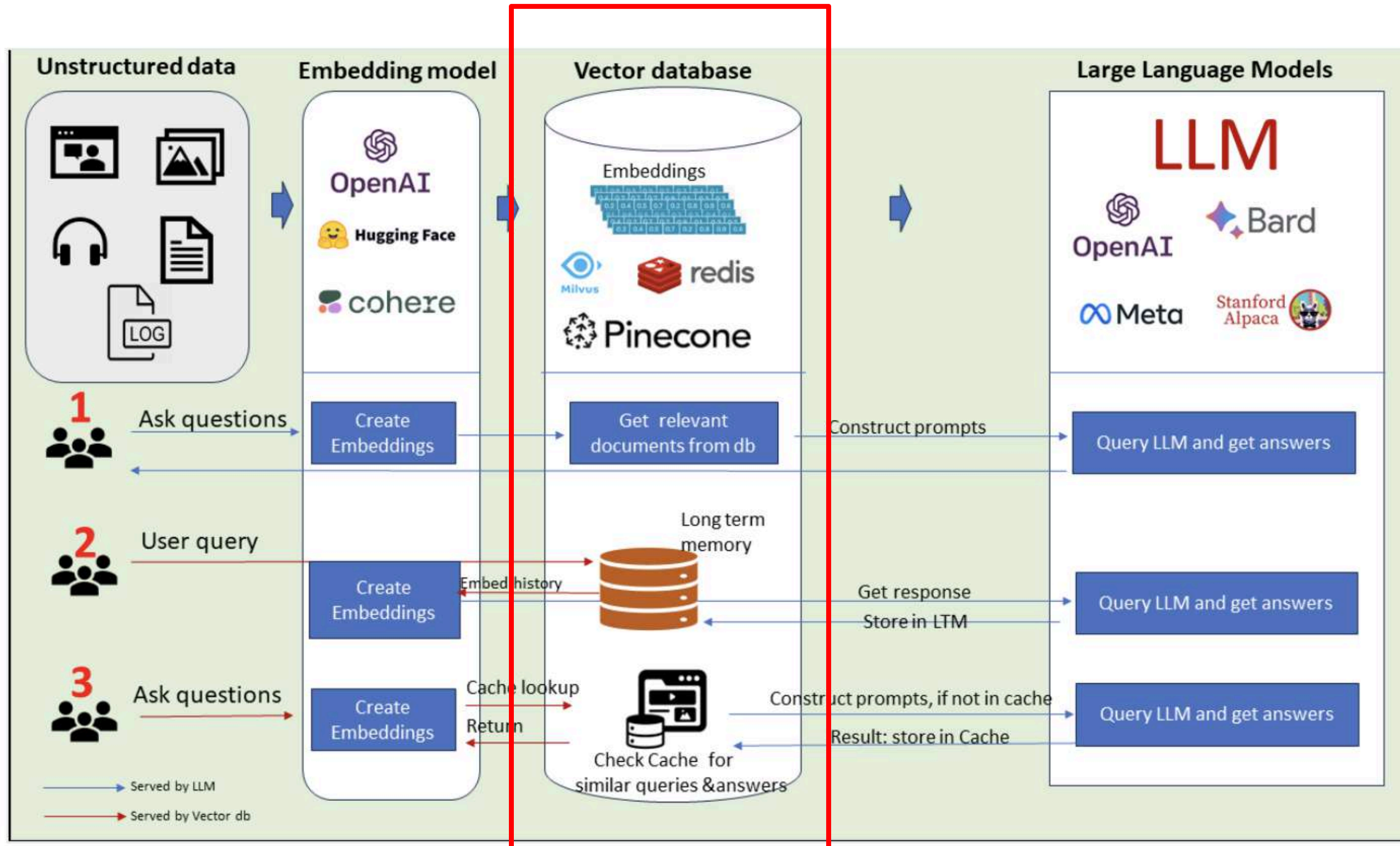
RAG (Retrieval-augmented generation)







Vector Database





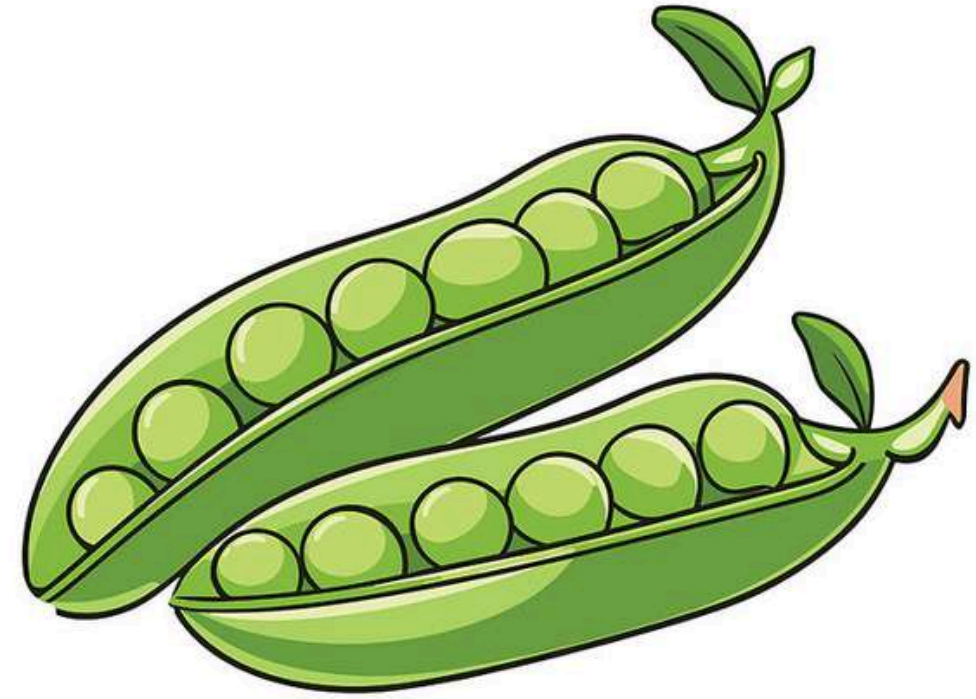
tea



coffee



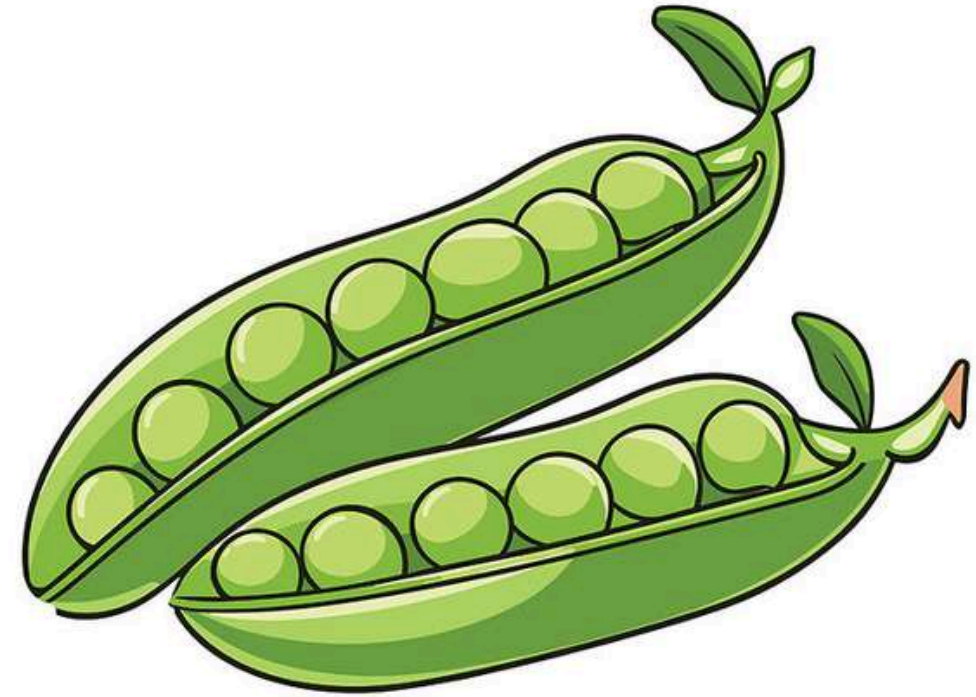
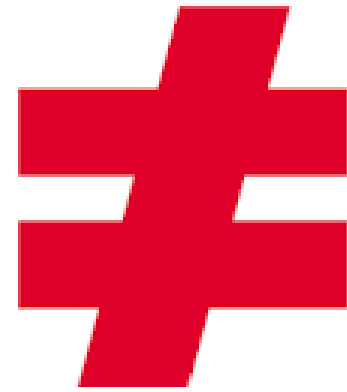
tea



pea



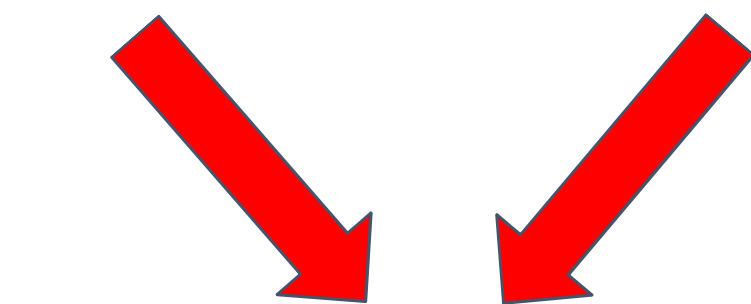
tea



pea



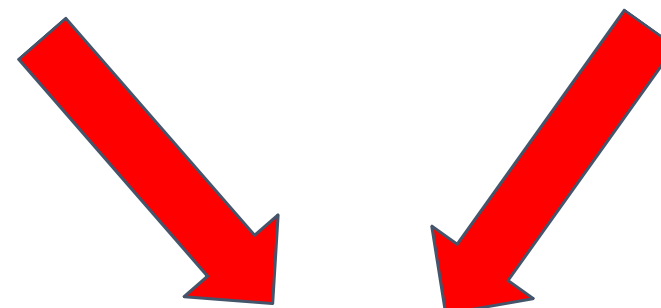
tea



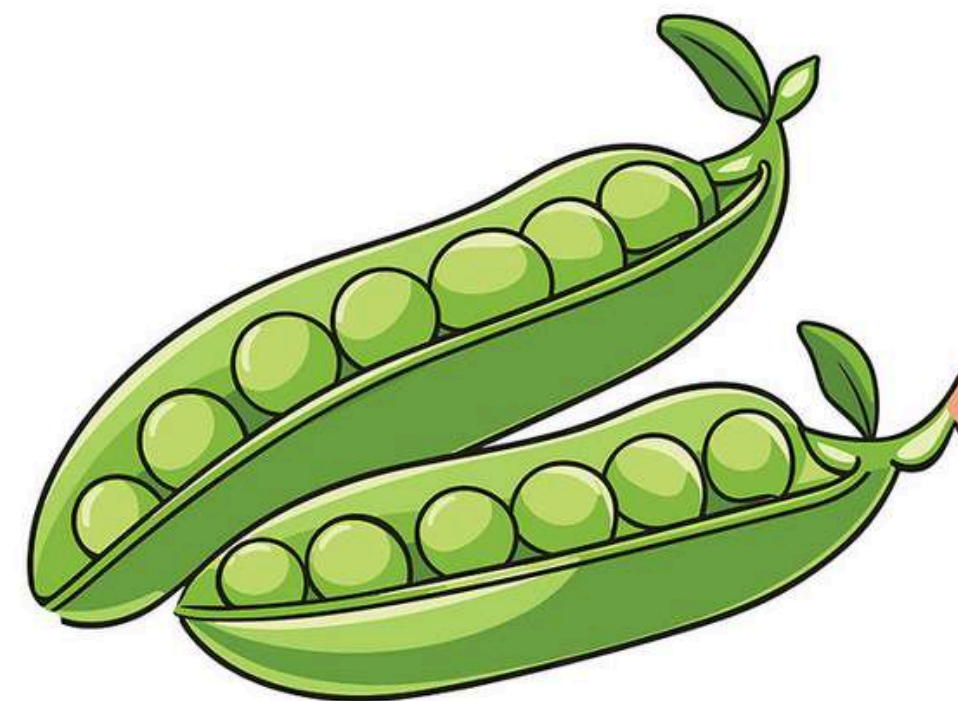
distance = 0.3



coffee

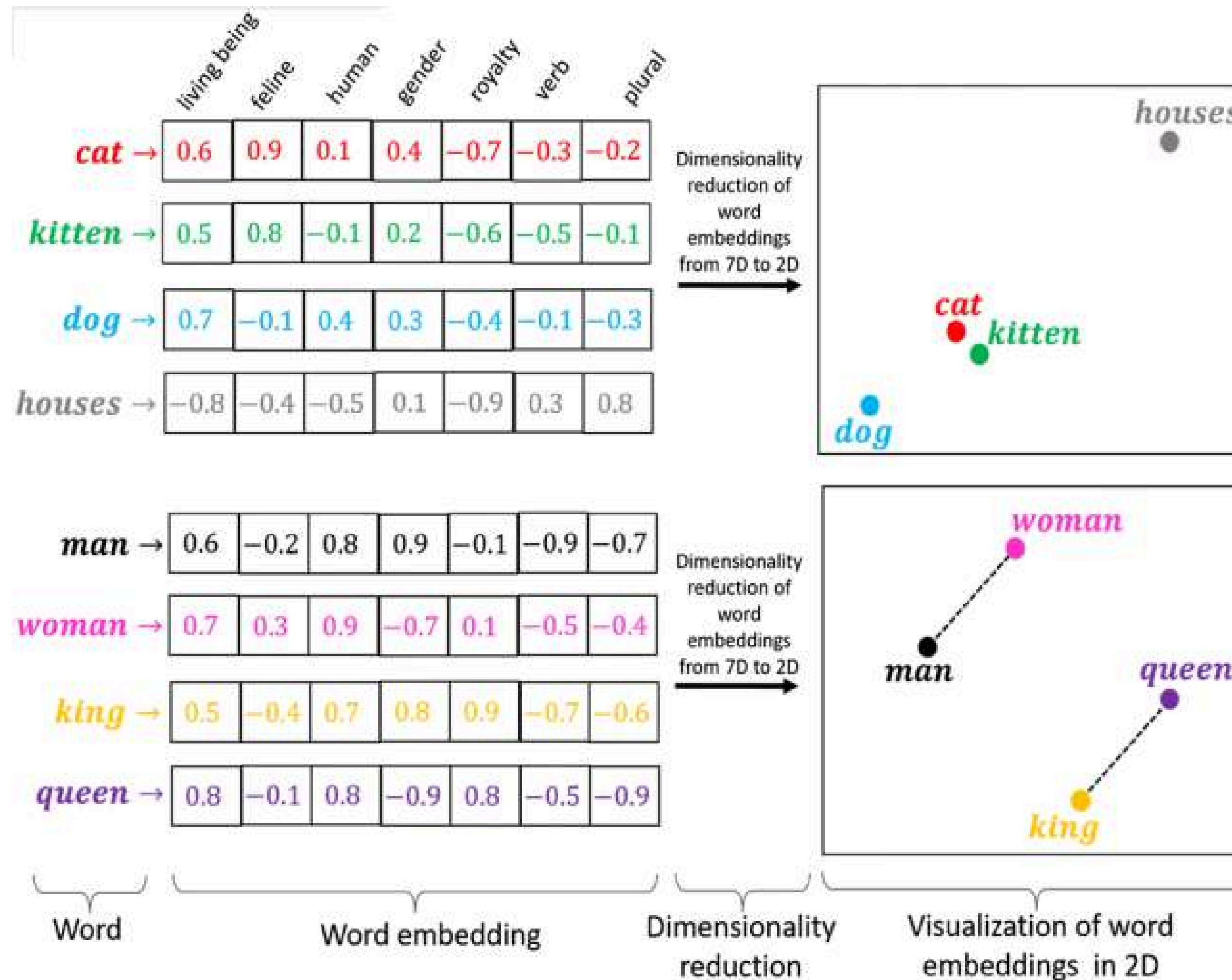


distance = 0.7

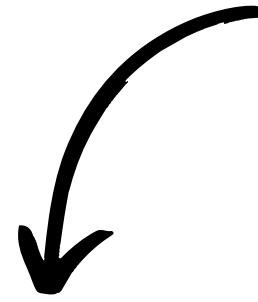


pea





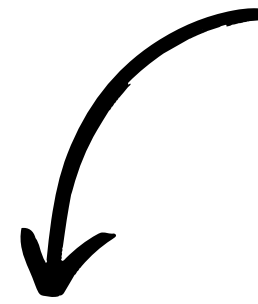
How can we get these word embedding vectors?



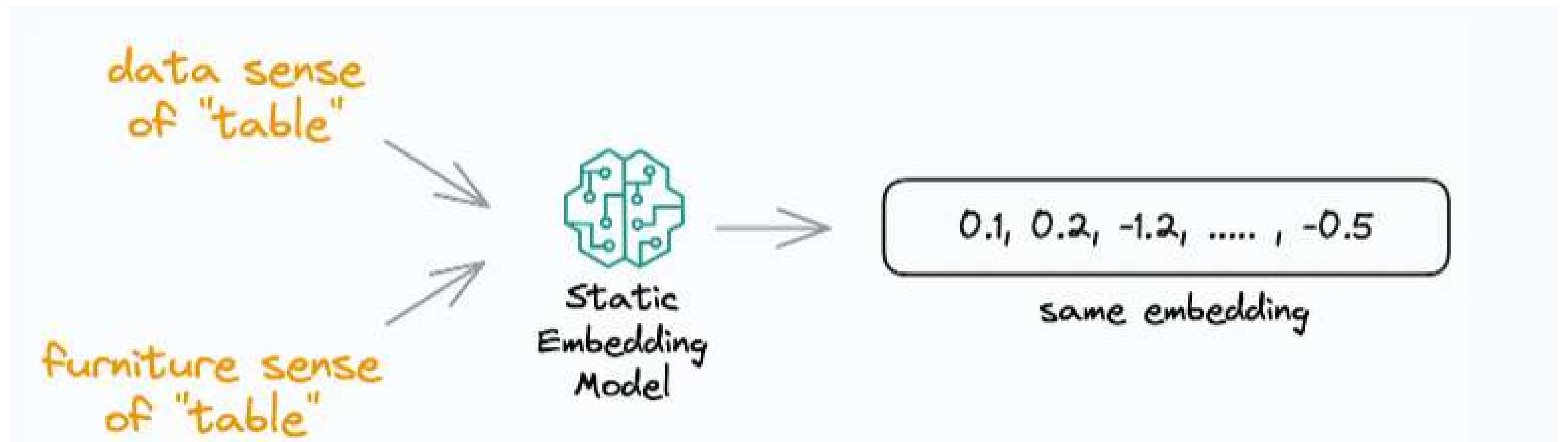
Pre-Trained Word Embeddings

- **Word2vec by Google 2013**
- **GloVe by Stanford
(Global Vectors for Word Representation)**
- **fasttext by Facebook**

Limitations



- Convert this data into a **table** in Excel.
- Put this bottle on the **table**.



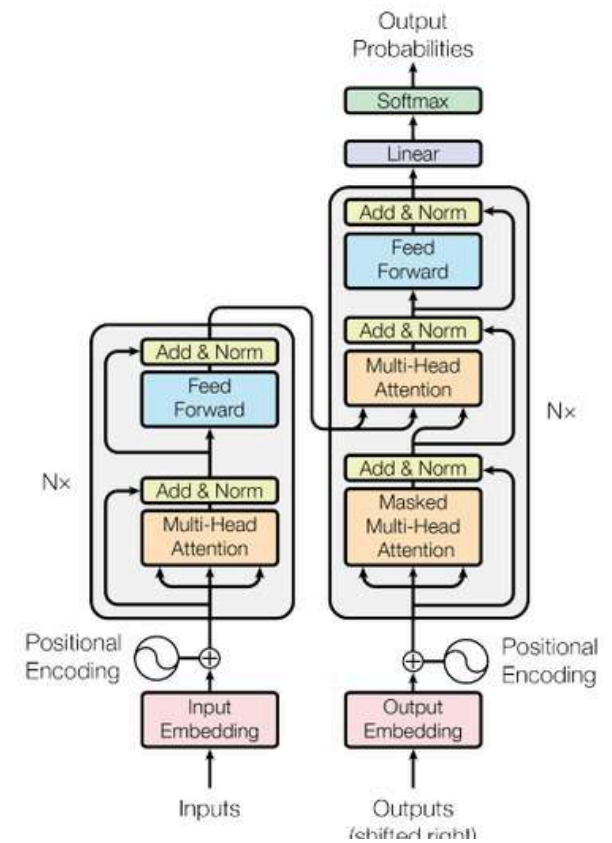
Solution ?

Transformer

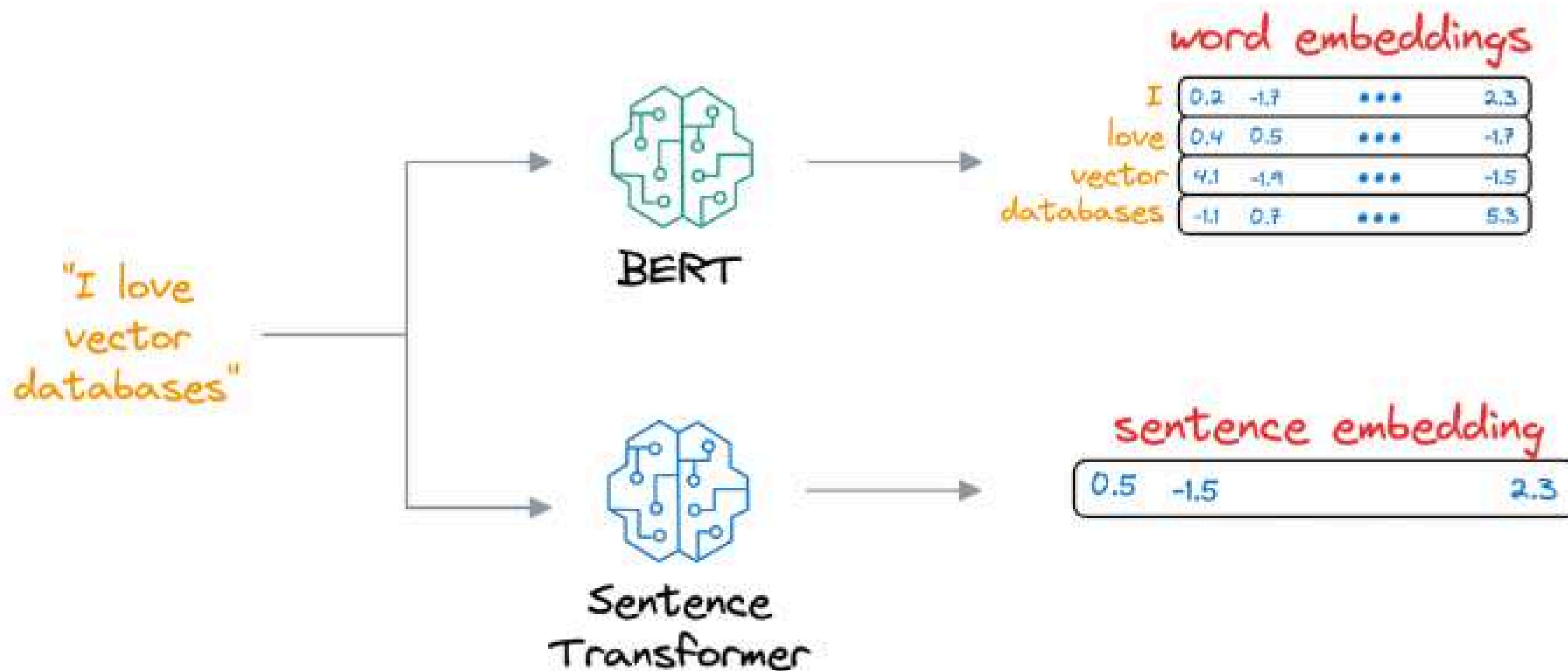
- BERT (Bidirectional Encoder Representations from Transformers)



- DistilBERT: BERT which is around 40% smaller:
- ALBERT: A Lite BERT (ALBERT).

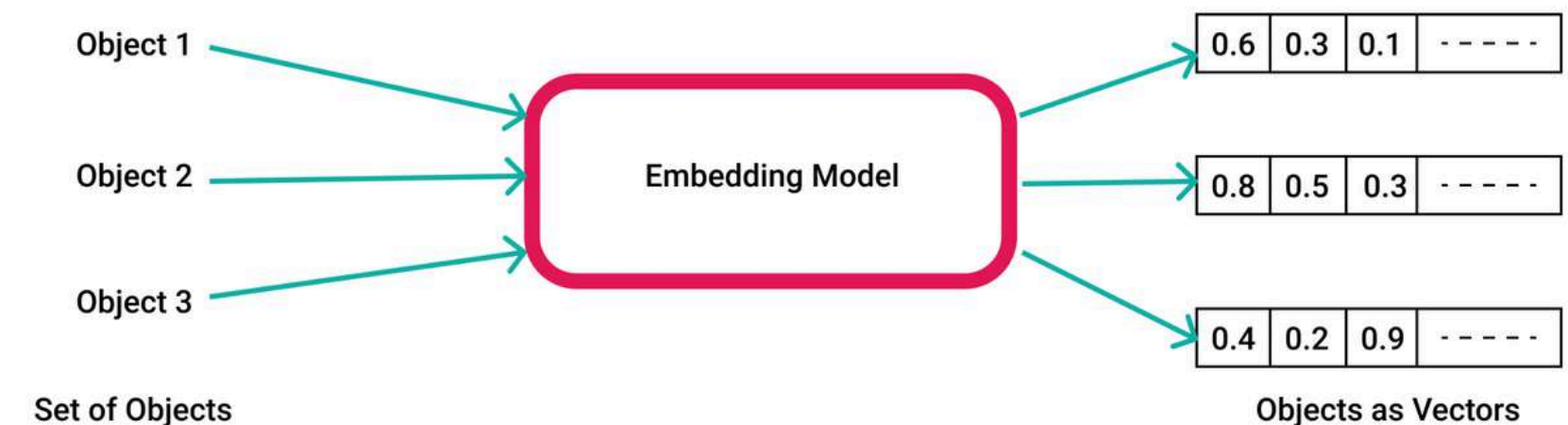


Solution ?

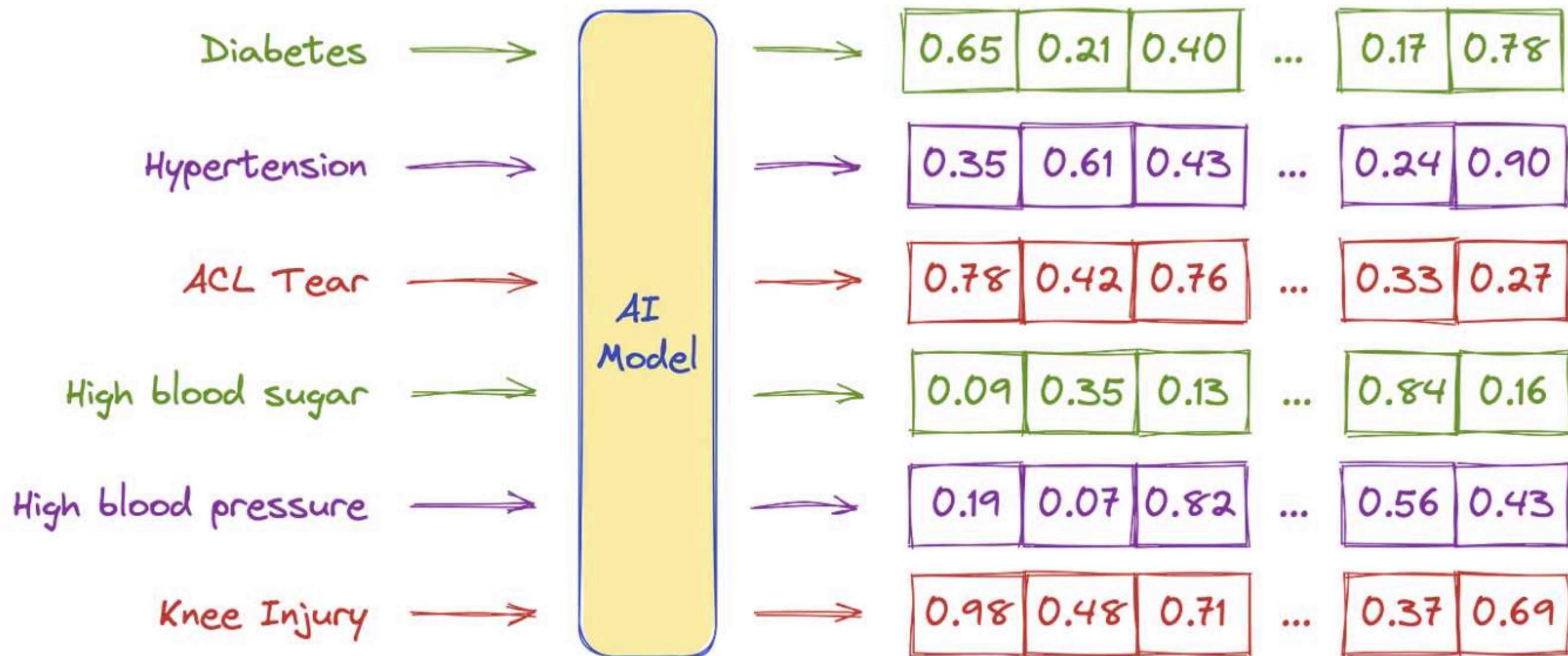


Embedding Model

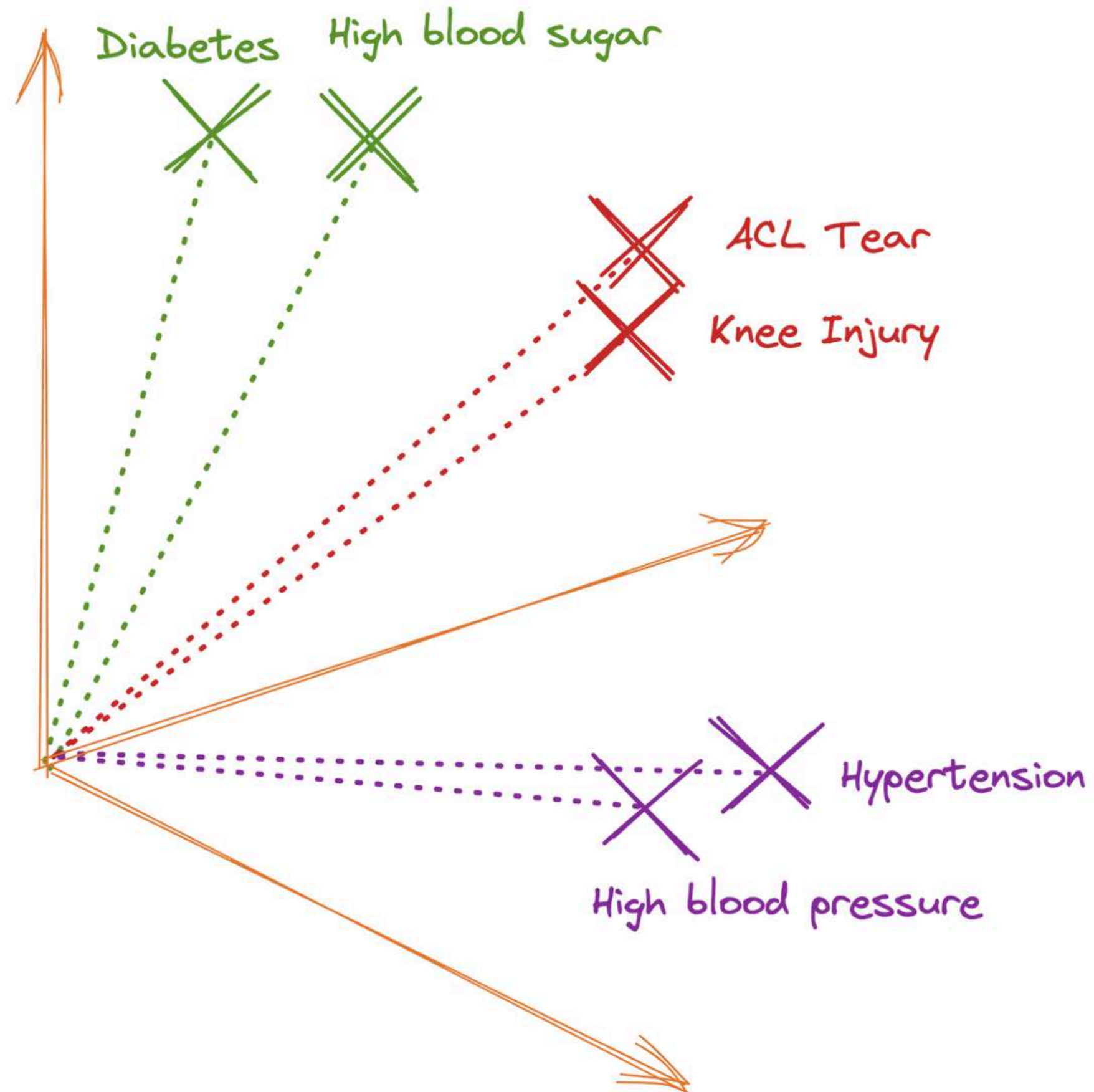
- These vectors live in a high-dimensional space where the proximity between vectors reflects the relatedness of the original items.
- **Embedding model** trained along LLM and learn to **produce representation(vectors) based on context** in word appear.



Embedding Space



Embedding Space



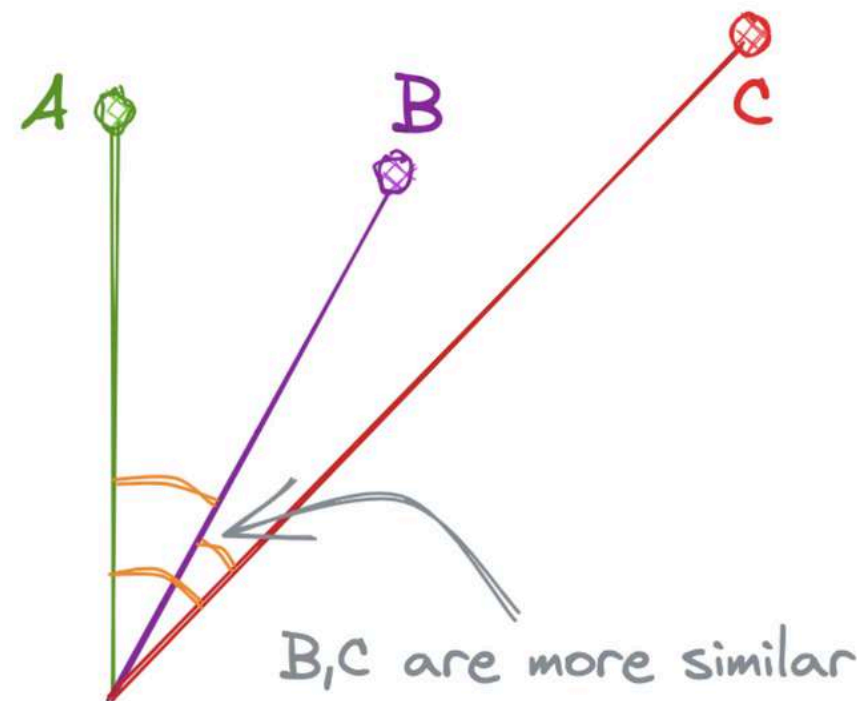
Querying a vector database

Similarity Calculation = > objective is to return the nearest neighbors

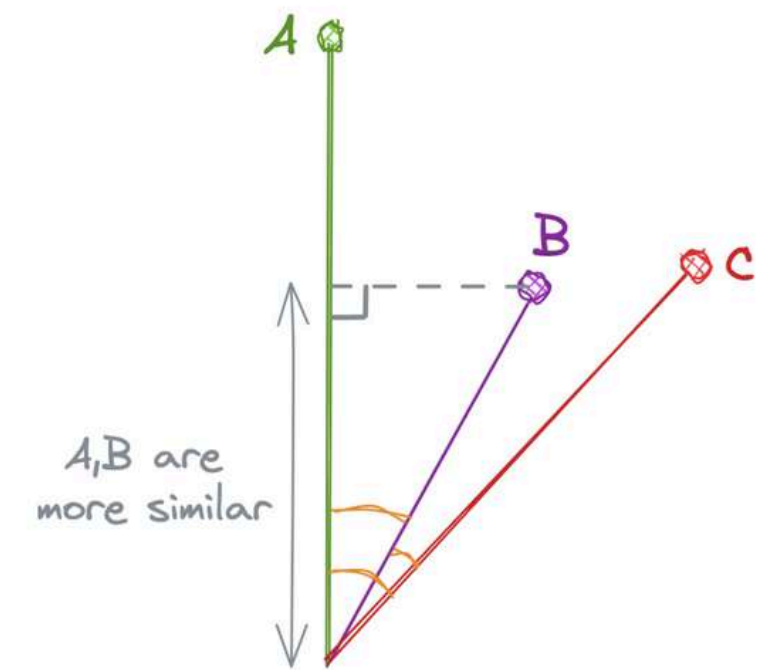
- For calculating similarity, there are several methods:
 - measuring distance - euclidean distance
 - cosine similarity or inner product



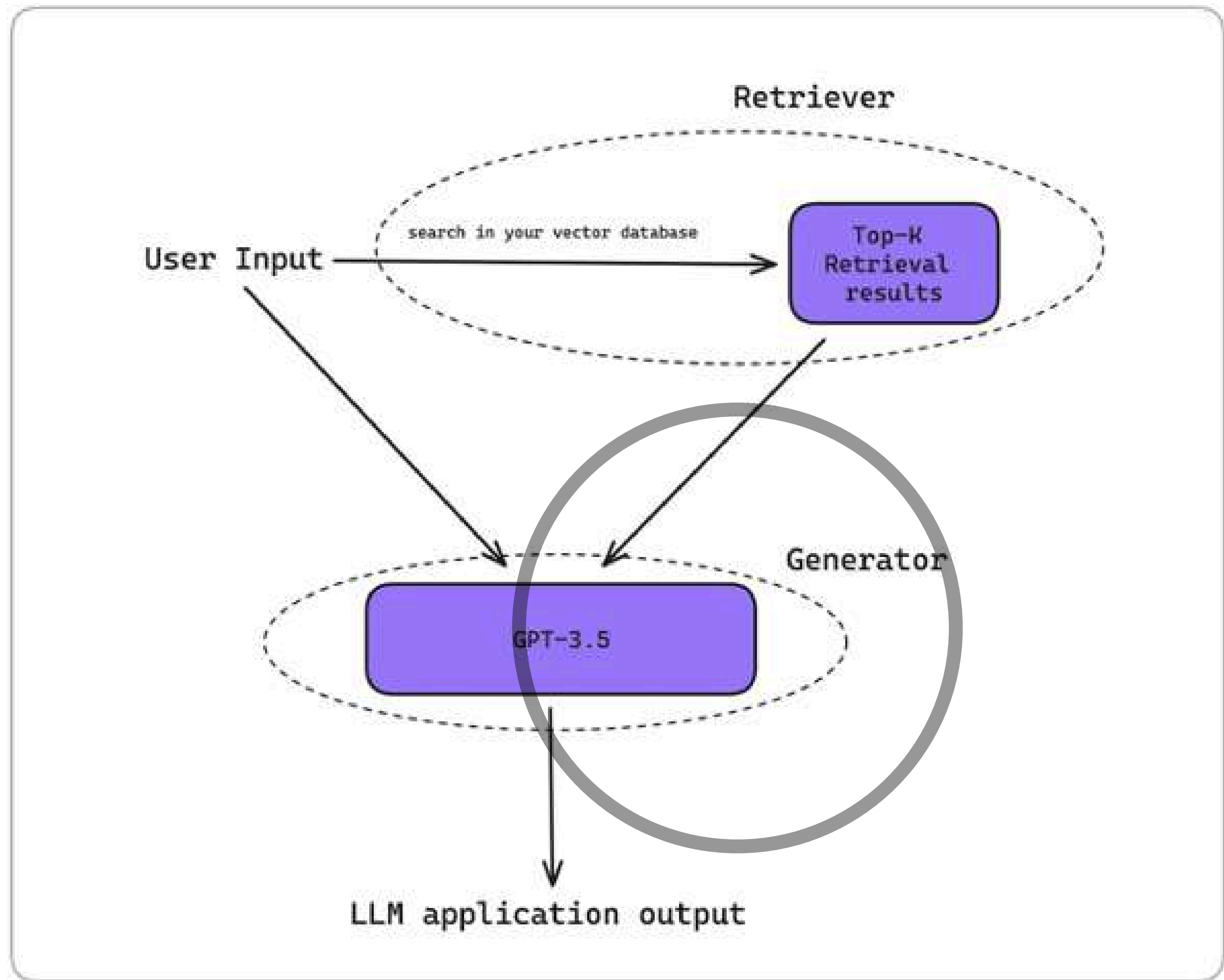
Euclidean distance

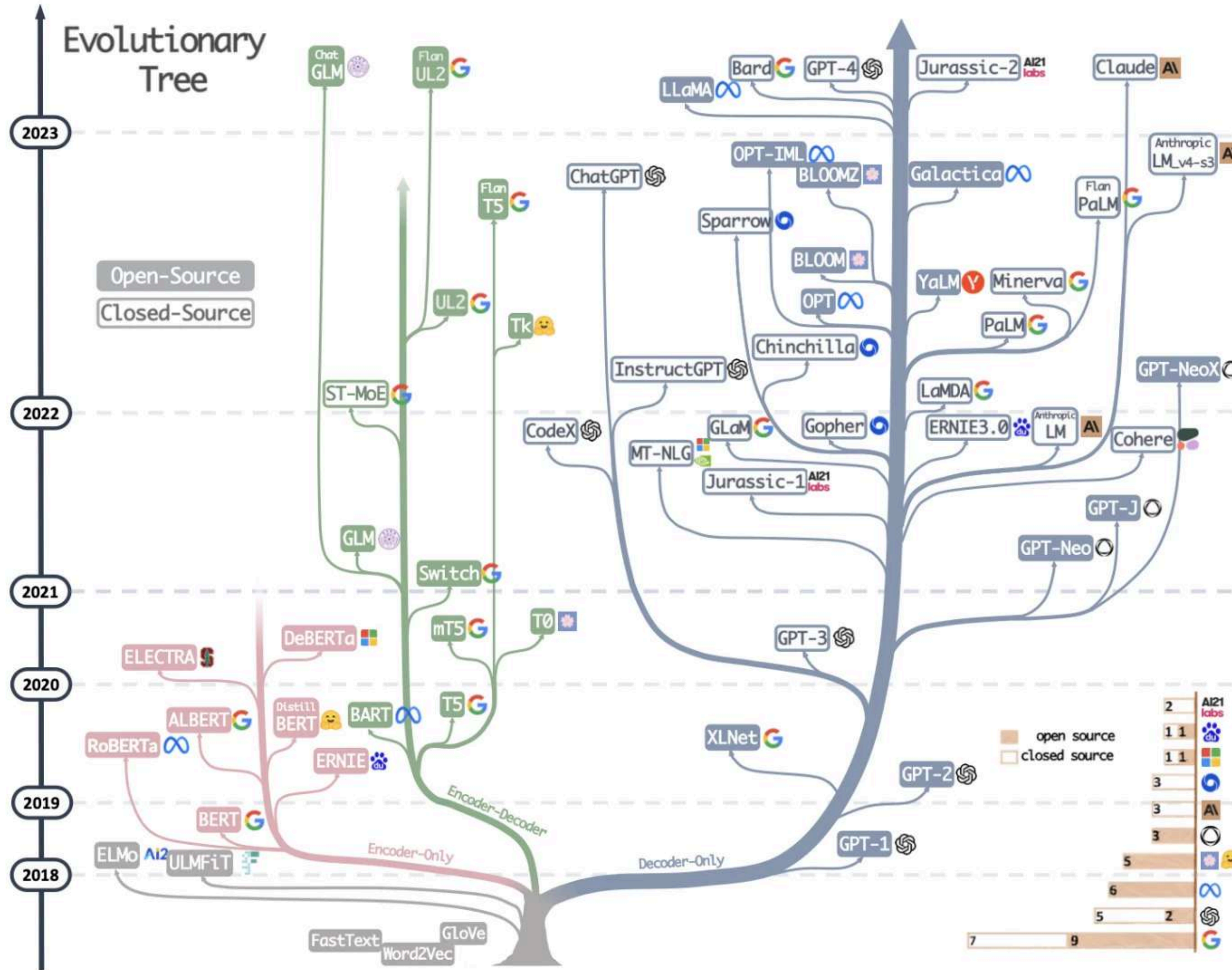


Cosine similarity



Dot Product





The evolutionary tree of modern LLMs via <https://arxiv.org/abs/2304.13712>.

Where to find pretrained LLMs ?



Hugging Face

By Mohammed Arbi Nsibi

Where to find pretrained LLMs ?

Models1,028,261

Filter by name

Full-text search

Sort: Trending

openai/whisper-large-v3-turbo

Automatic Speech Recognition • Updated 1 day ago • 10k • 324

black-forest-labs/FLUX.1-dev

Text-to-Image • Updated Aug 16 • 1.14M • 5.03k

jasperai/Flux.1-dev-Controlnet-Upscaler

Image-to-Image • Updated 3 days ago • 9.86k • 244

allenai/Molmo-7B-D-0924

Image-Text-to-Text • Updated 1 day ago • 14.5k • 273

meta-llama/Llama-3.2-11B-Vision-Instruct

Image-Text-to-Text • Updated 4 days ago • 139k • 479

nvidia/NVLM-D-72B

Image-Text-to-Text • Updated about 18 hours ago • 860 • 242

meta-llama/Llama-3.2-1B

Text Generation • Updated 3 days ago • 61.2k • 299

openbmb/MiniCPM-Embedding

Feature Extraction • Updated 2 days ago • 130k • 204

Datasets222,500

Filter by name

Full-text search

Sort: Trending

google/frames-benchmark

Viewer • Updated about 17 hours ago • 824 • 562 • 122

FBK-MT/mosel

Viewer • Updated 5 days ago • 51.1M • 21 • 42

openai/MMMLU

Viewer • Updated 4 days ago • 393k • 5.33k • 374

argilla/FinePersonas-v0.1

Viewer • Updated 19 days ago • 21.1M • 371 • 304

fka/awesome-chatgpt-prompts

Viewer • Updated Sep 3 • 170 • 8.36k • 5.82k

migtissera/Synthia-v1.5-I

Viewer • Updated 8 days ago • 20.7k • 99 • 39

HackerNoon/where-startups-trend

Preview • Updated 7 days ago • 19 • 36

k-mktr/improved-flux-prompts-photoreal-portrait

Viewer • Updated 4 days ago • 20k • 54 • 62

Image-Text-to-Text

Visual Question Answering

Document Question Answering

Video-Text-to-Text

Any-to-Any

Computer Vision

Depth Estimation

Image Classification

Object Detection

Image Segmentation

Text-to-Image

Image-to-Text

Image-to-Image

Image-to-Video

Unconditional Image Generation

Video Classification

Text-to-Video

Zero-Shot Image Classification

Mask Generation

Zero-Shot Object Detection

Text-to-3D

Image-to-3D

Image Feature Extraction

Keypoint Detection

Natural Language Processing

Text Classification

Token Classification

Table Question Answering

Question Answering

Zero-Shot Classification

Translation

Summarization

Feature Extraction

Text Generation

Text2Text Generation

Fill-Mask

Sentence Similarity

Audio

Text-to-Speech

Text-to-Audio

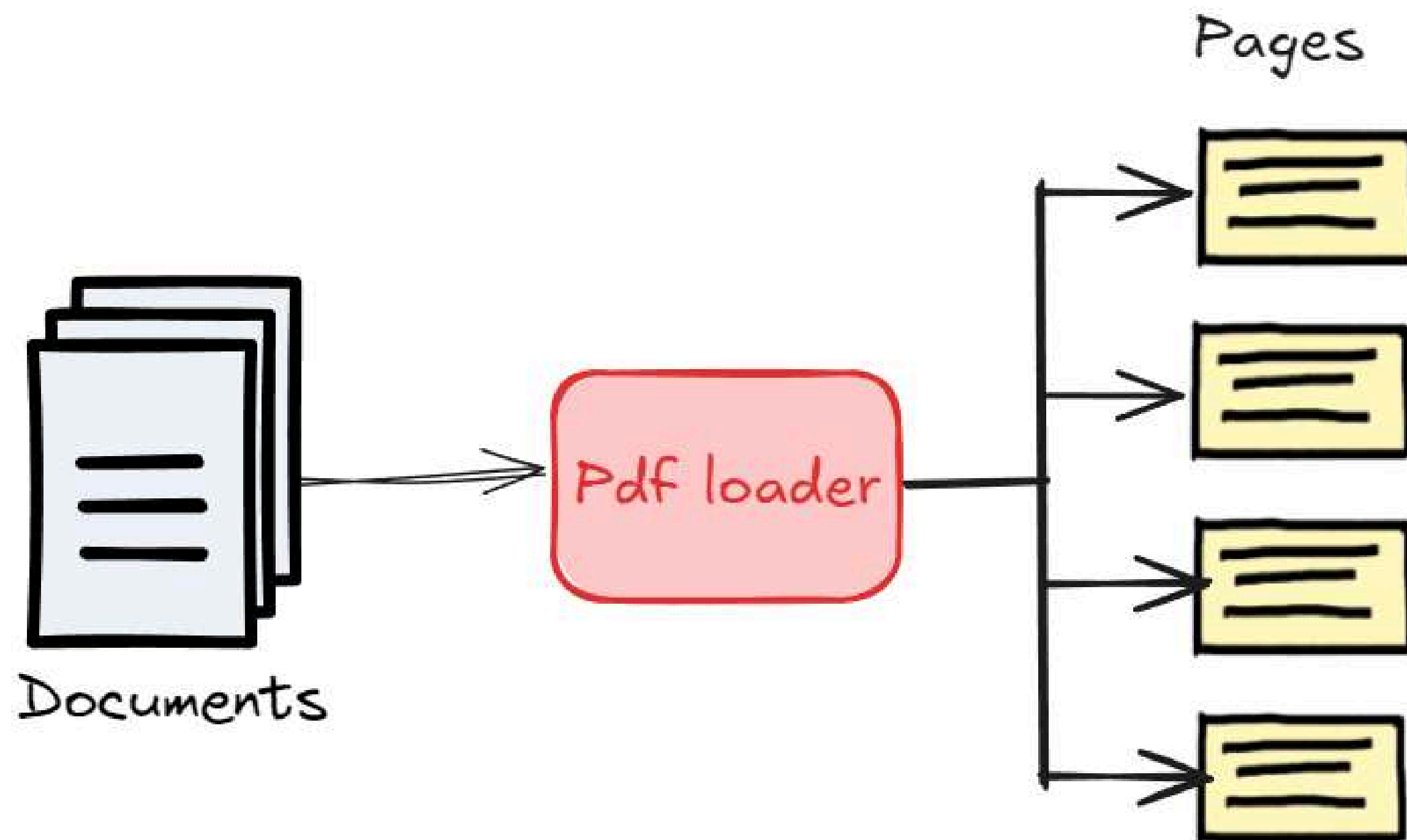
Automatic Speech Recognition

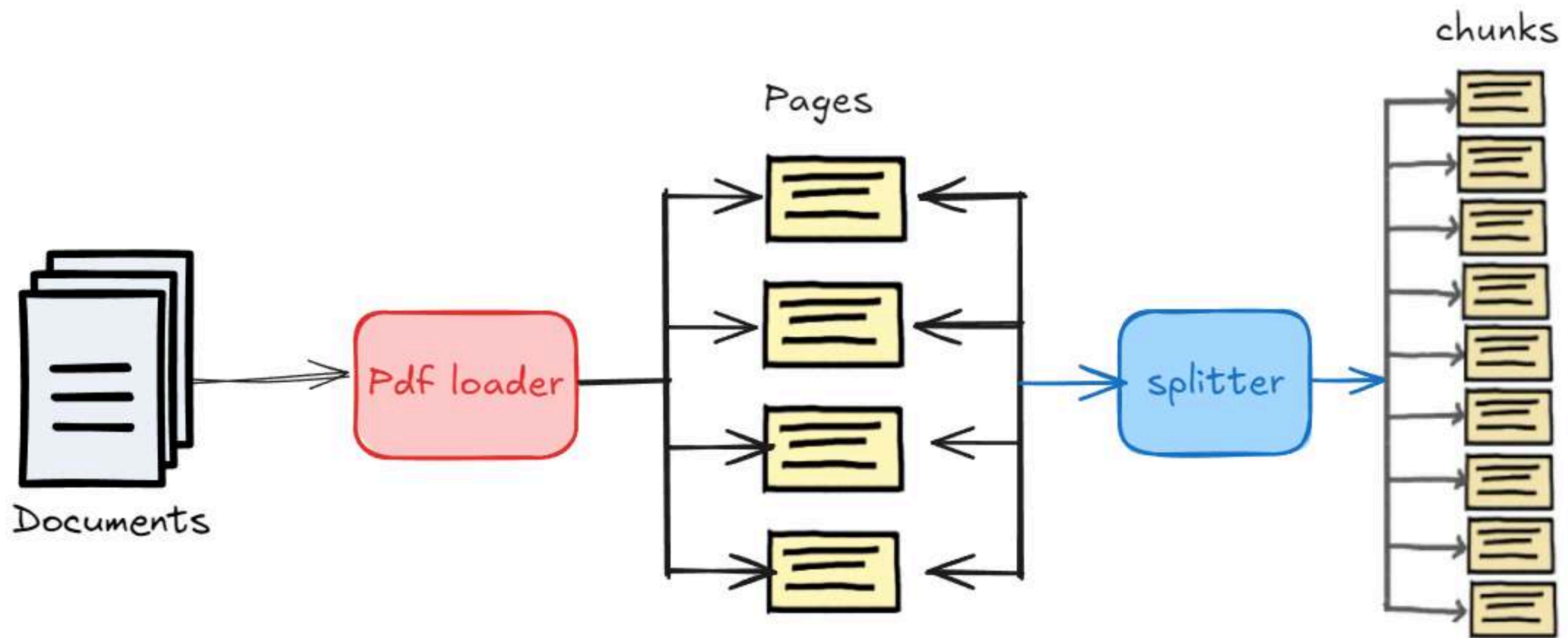
Audio-to-Audio

Audio Classification

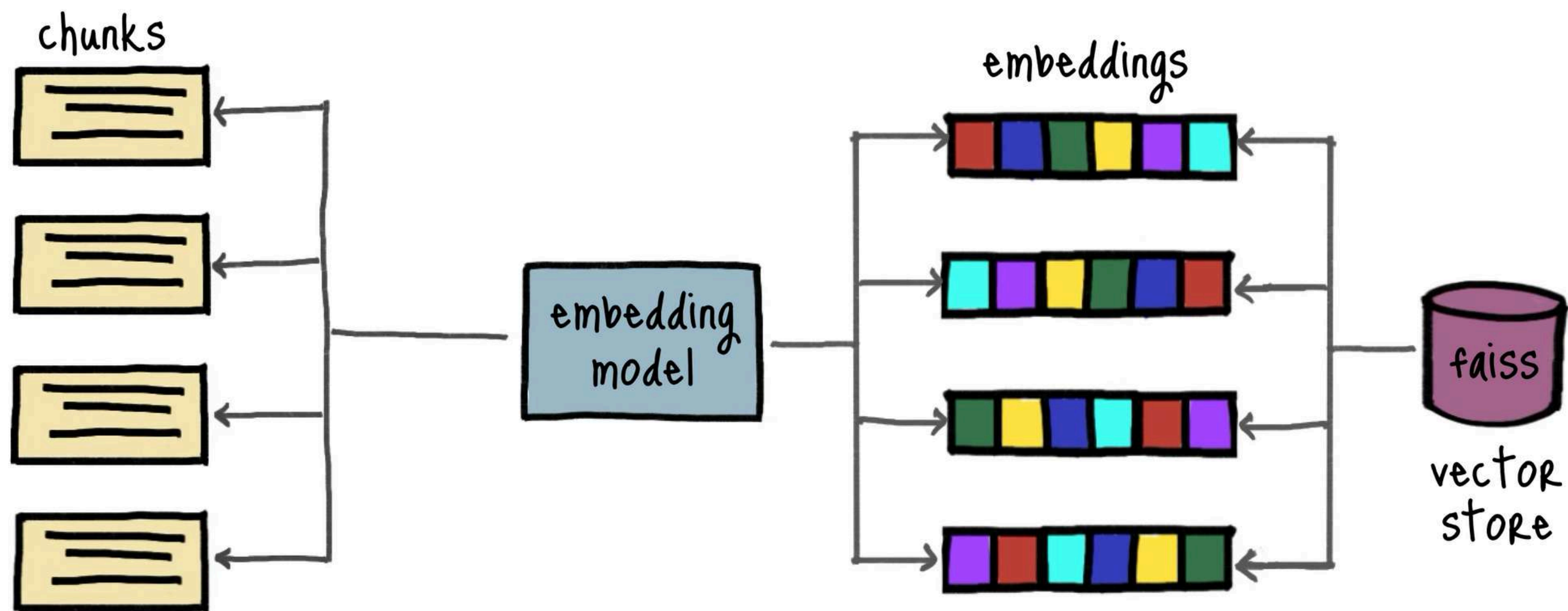
Voice Activity Detection

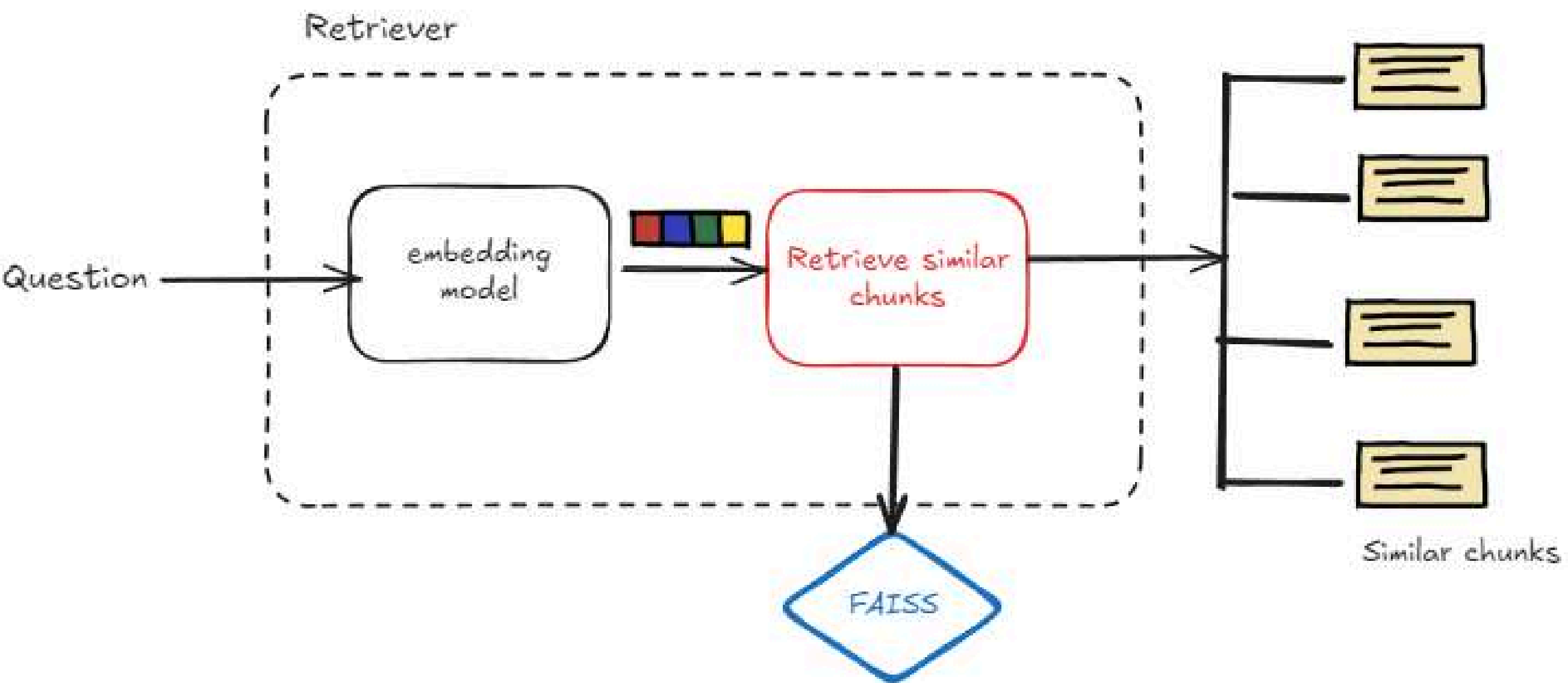
RAG architecture



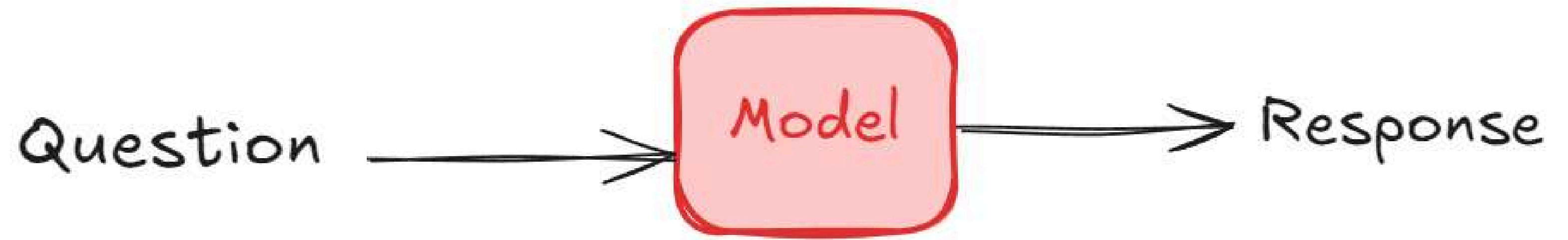


By Mohammed Arbi Nsibi

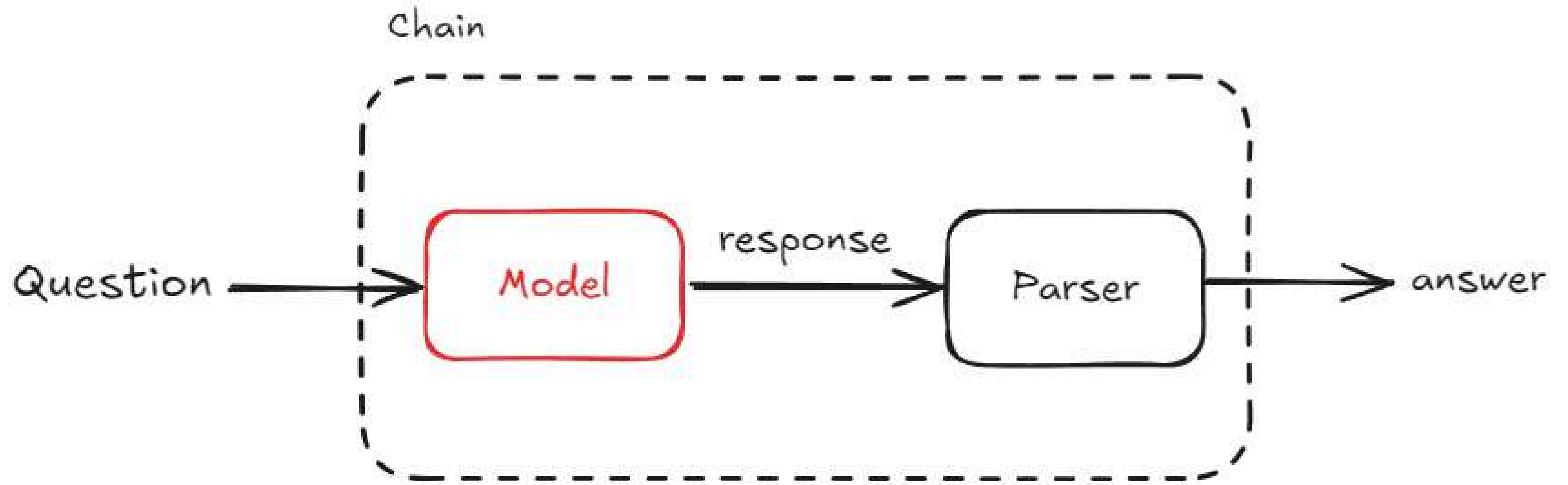




By Mohammed Arbi Nsibi




```
AIMessage(content='As of my last update in April 2023, Joe Biden is the President of the United States. He took office on January 20, 2021,
```

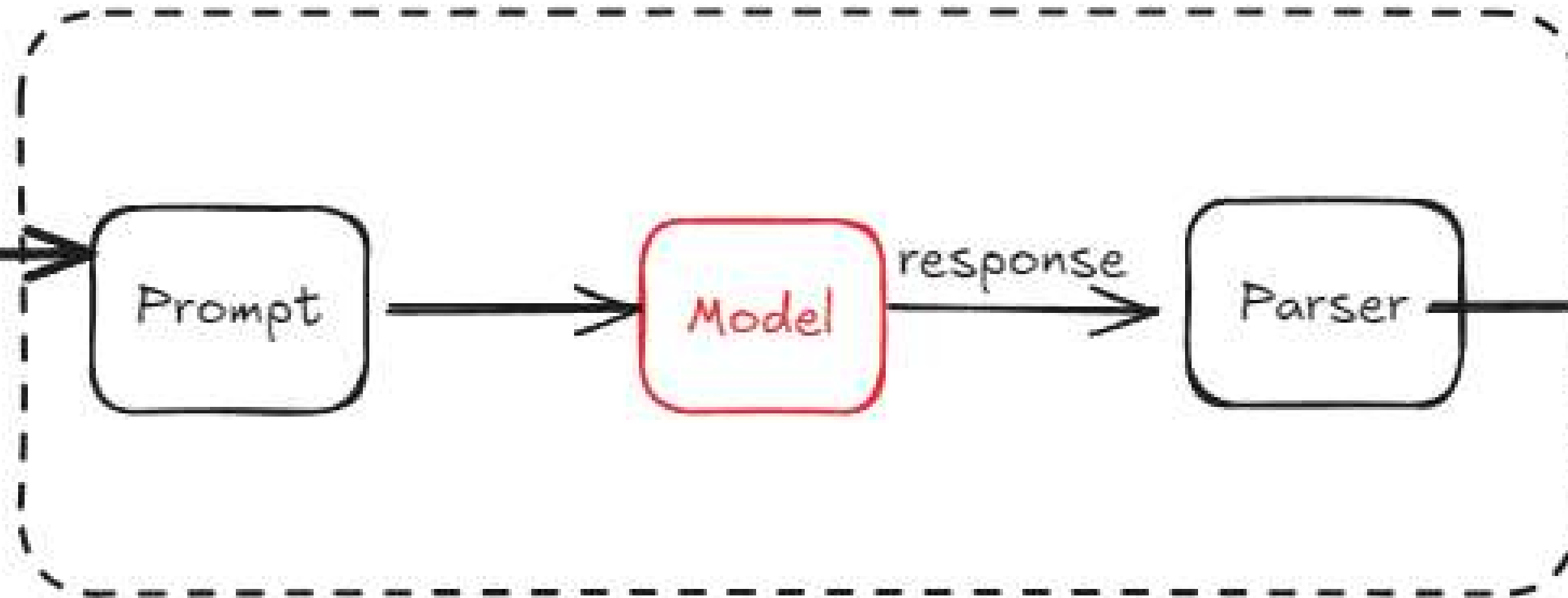


```
'As of my last update in April 2023, Joe Biden is the President of the United States. He took office on January 20, 2021,
```

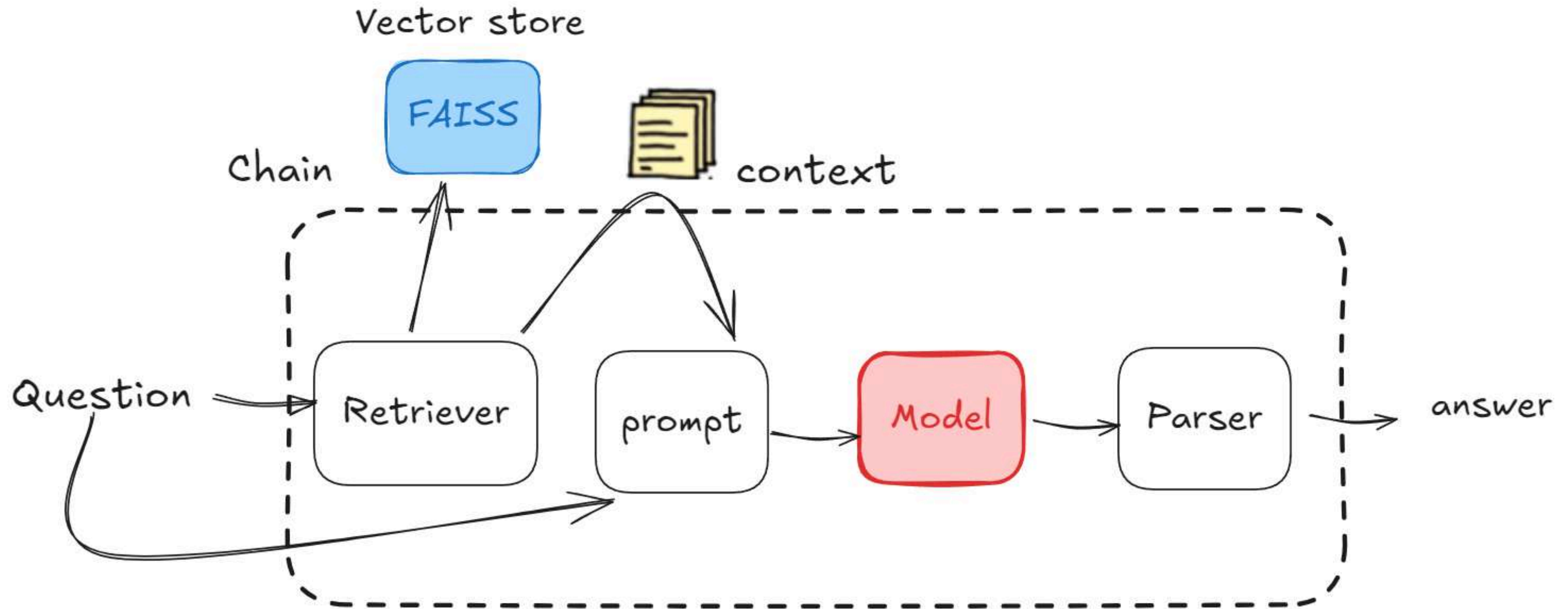


Chain

context
+
Question



By Mohammed Arbi Nsibi



How to get started ?

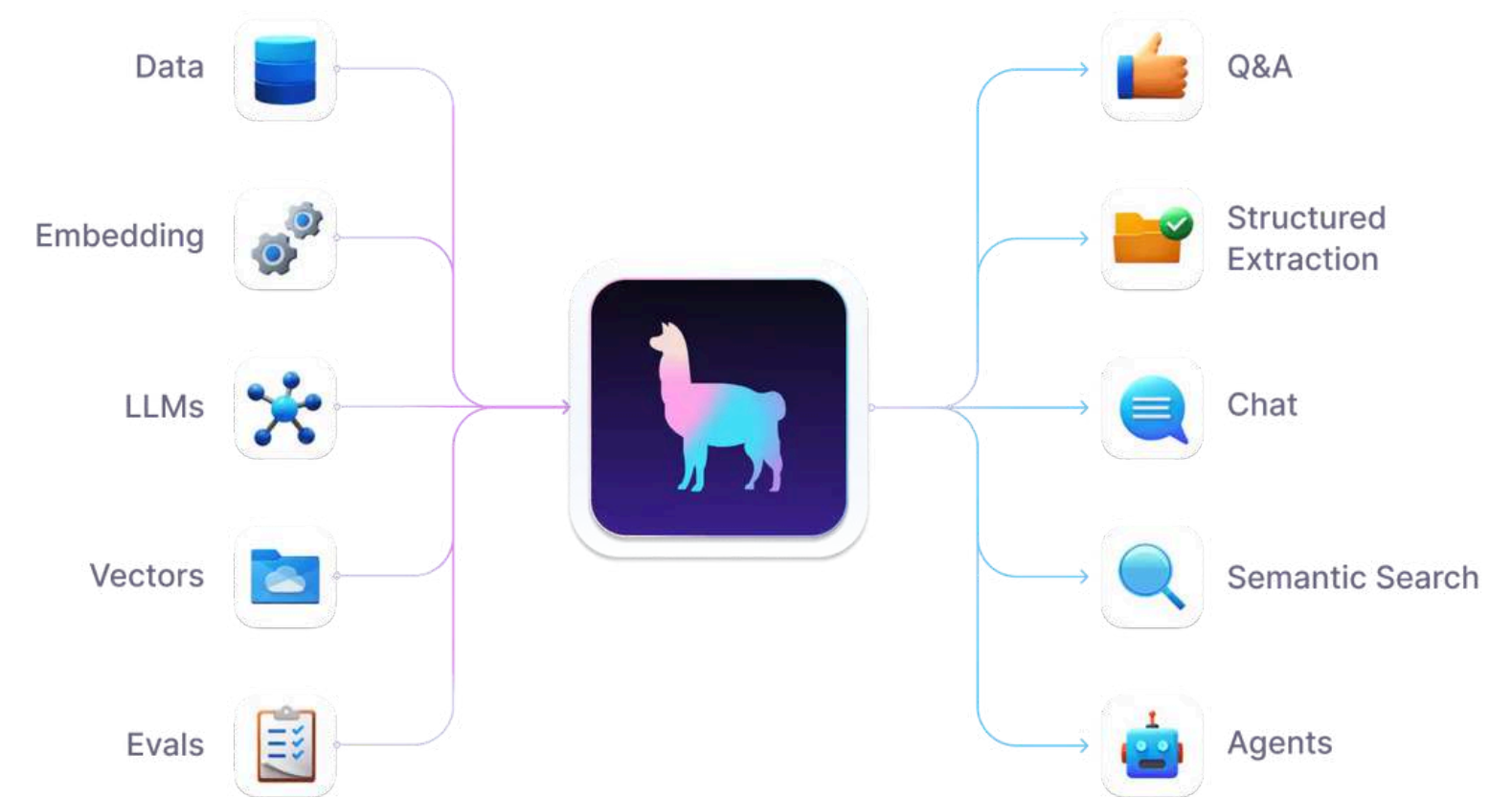
LangChain

- [LangChain](#) is a framework designed to simplify the creation of applications using large language models.
- It is based on LCEL (LangChain Expression Language)
- Use-cases including chatbots, RAG, document summarization and synthetic data generation.



LlamaIndex

- [LlamaIndex](#) is a handy tool that acts as a bridge between your custom data and large language models (LLMs) which are powerful models capable of understanding human-like text.
- Since majority applications are RAG, LlamaIndex provides the right tools to build RAG



LET'S CODE 🐱



By Mohammed Arbi Nsibi

always has been

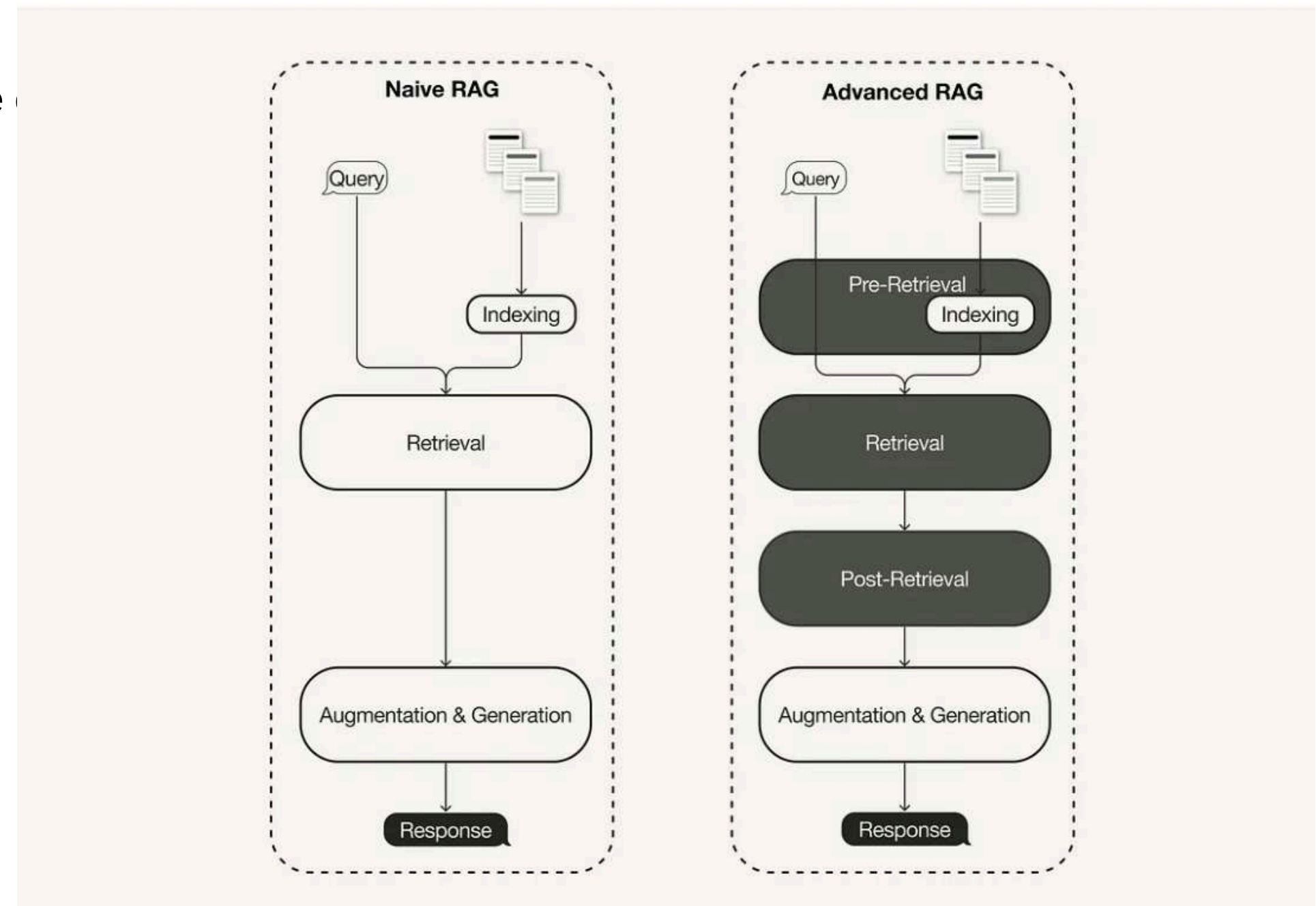
wait so creating chatbots is that easy ?





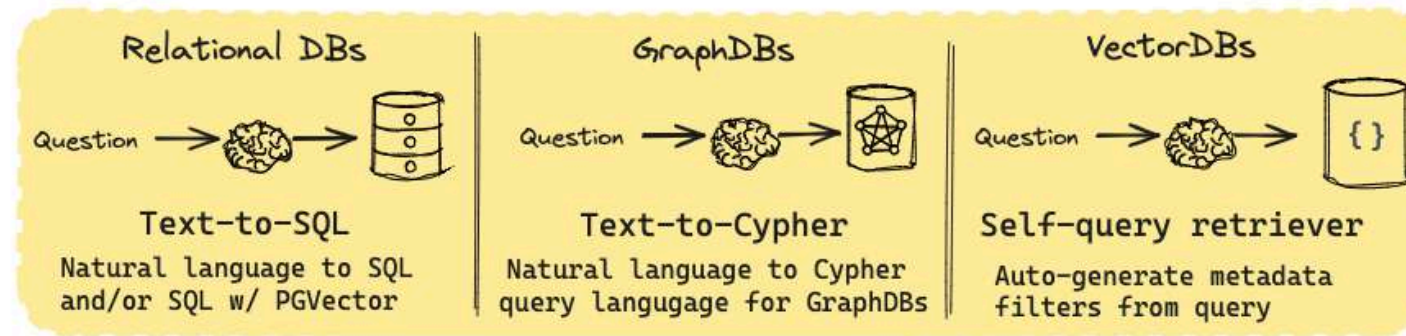
Naive RAG vs Advanced RAG

- There are many implementation to further improve performance Naive RAG.
- Advanced RAG has evolved as a new paradigm with targeted enhancements to address some of the limitations of the naive RAG paradigm.
- Advanced RAG techniques can be categorized into
 - pre-retrieval optimization,
 - retrieval optimization, and
 - post-retrieval optimization

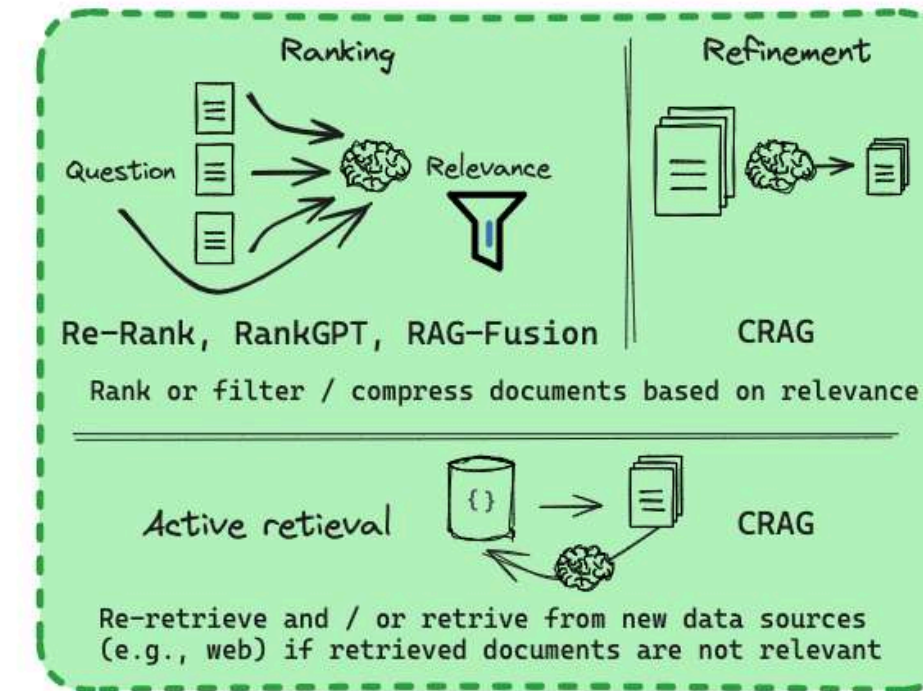


Difference between Naive and Advanced RAG (Image by the author, inspired by [1])

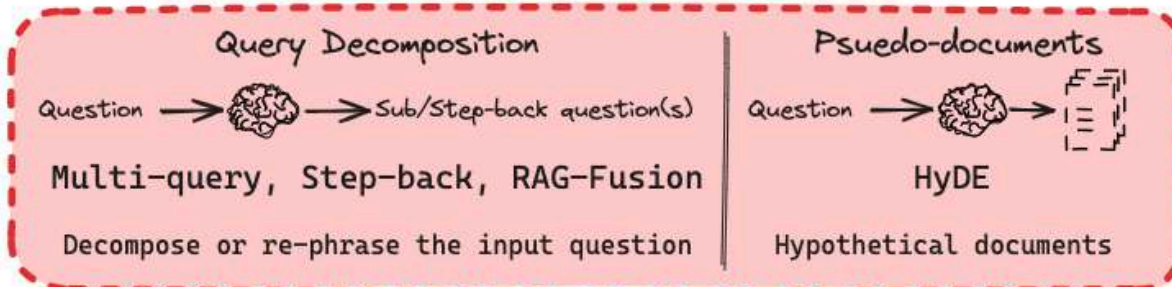
Query Construction



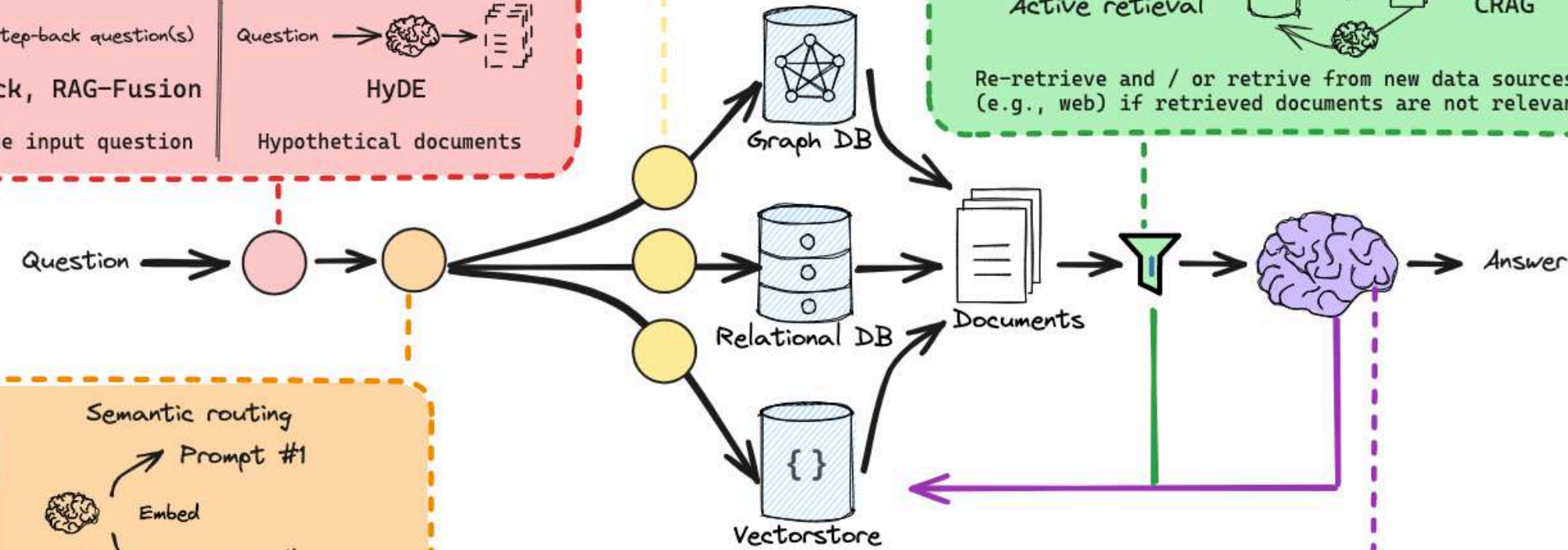
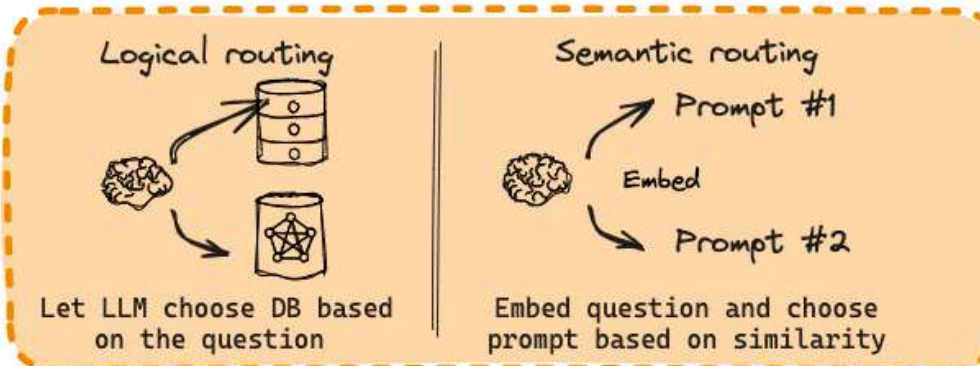
Retrieval



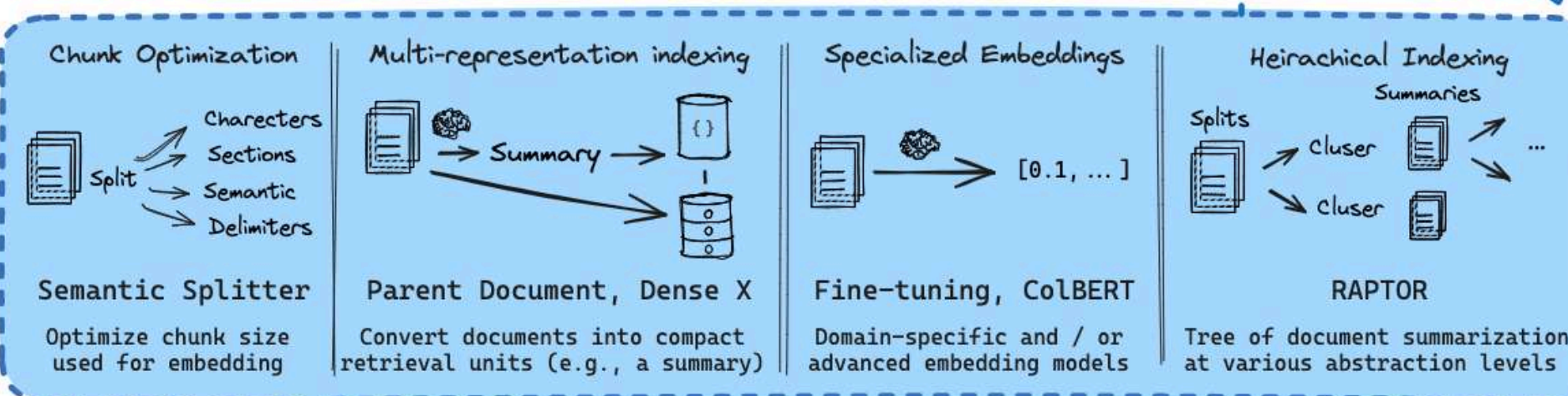
Query Translation



Routing



Indexing



Generation





Resources

- [Your RAG powered by Google Search Technology.](#)
- [Let's talk about LlamaIndex and LangChain](#)
- [Retrieval-Augmented Generation \(RAG\) framework in Generative AI](#)
- [Retrieval-Augmented Generation \(RAG\): From Theory to LangChain Implementation](#)
- [Advanced Retrieval-Augmented Generation: From Theory to LlamaIndex Implementation](#)
- [Retrieval-augmented generation for large language models: A survey.](#)
[\[arXiv\]](#)

Q&A

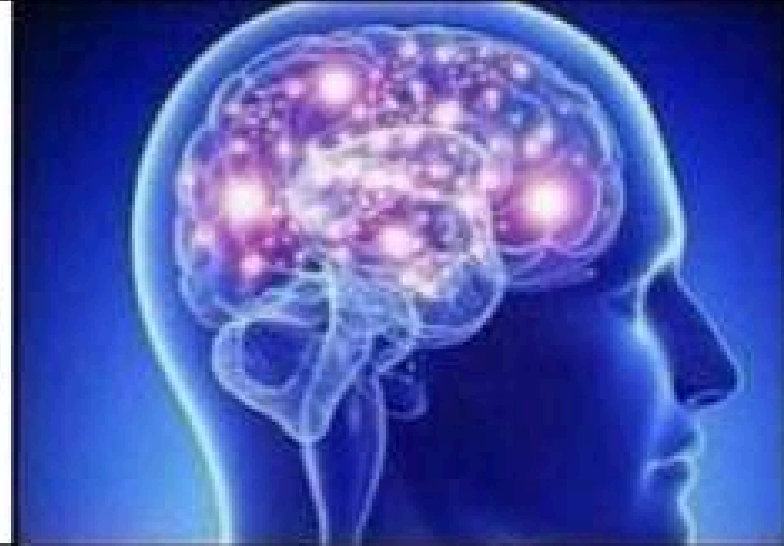
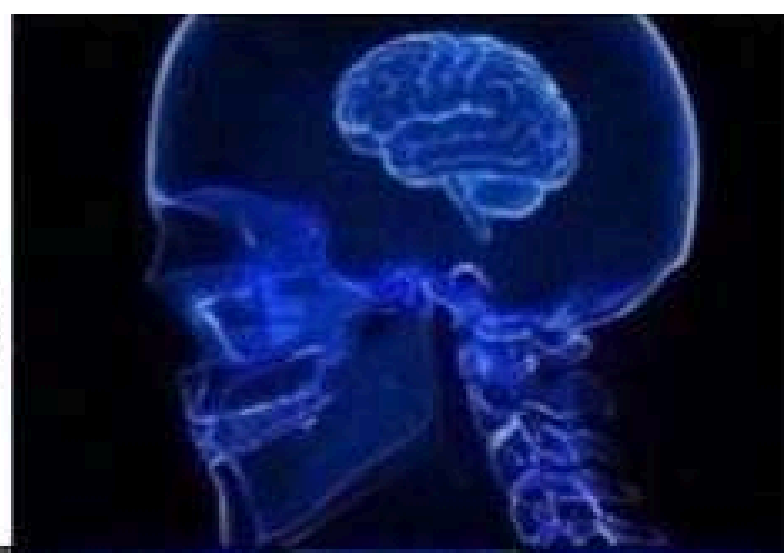
THANK YOU for you attention!!

**USING AI
FOR CALCULATIONS**

**USING AI TO
WRITE ESSAYS**

**USING AI TO
GENERATE CODE**

**USING AI
TO GENERATE
MEMES ABOUT AI**



QUIZ TIME

